DS4023 (2025 Fall)

Machine Learning

Department of Computer Science
BNU-HKBU United International College

Course Information

- Teaching:
  - Lecture: Dr. Simon LUO
  - Office: T3-501-R3, Email: zongweiluo@uic.edu.cn
  - Teaching Assistant:
    - Office: TBA, Email:
- Lecture Scheduling:
  - Tue 10:00-10:50 (T29-401), Thu 11:00-11:50 (T7-503)

Simon LUO

- Office: T3-502-R3
- Office hours: Tue 12:00-13:00, Thu 9:00-11:00, 12:00-13:00
- Research area: Causal inference, LLM Prompting, FinTech

Assessment Method

- Assignment (Writing and Programming): 30%
- Final Project: 30%
- Final Exam (One cheat page): 40%

Schedule

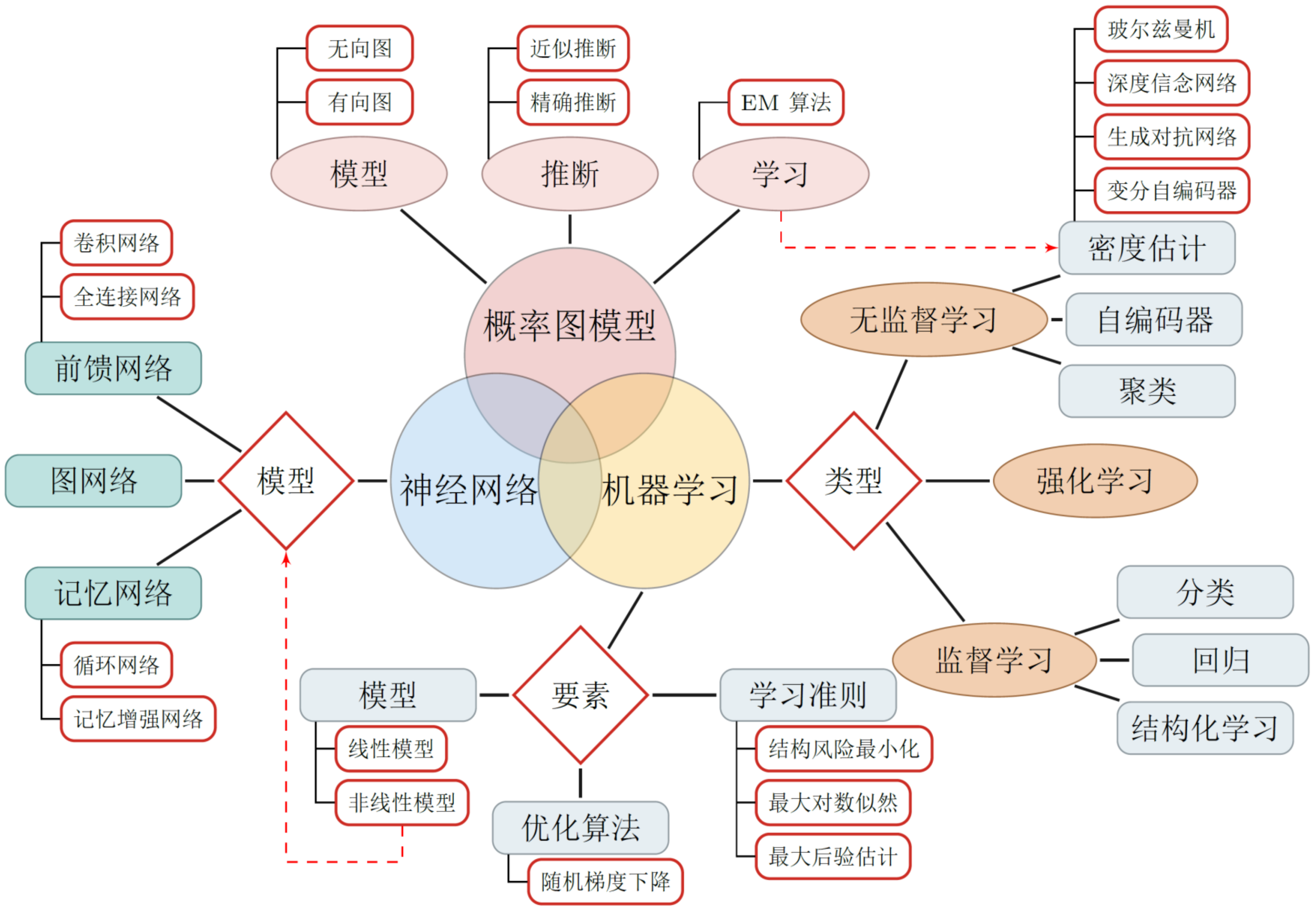
Week	Teaching content		Assignment
1	Introduction		
2	Statistical learning	Parameter estimation, Bayes	A1
3		MAP, MPD	
4		Linear regression	A2
5	Supervised learning	Logistic regression, SVM	
6		Boosting, random forest	A3
7		Neural networks	Project
8	Reading week		
9		Deep learning	A4
10	Unsupervised learning	K-means, HC, GMM, EM	
11		Self organizing map, KDE,	
12	Dimensionality reduction	Subspace method, non-linear manifold embedding	A5
13	Final PRE		
14	Final PRE		

Chapter 1: Introduction

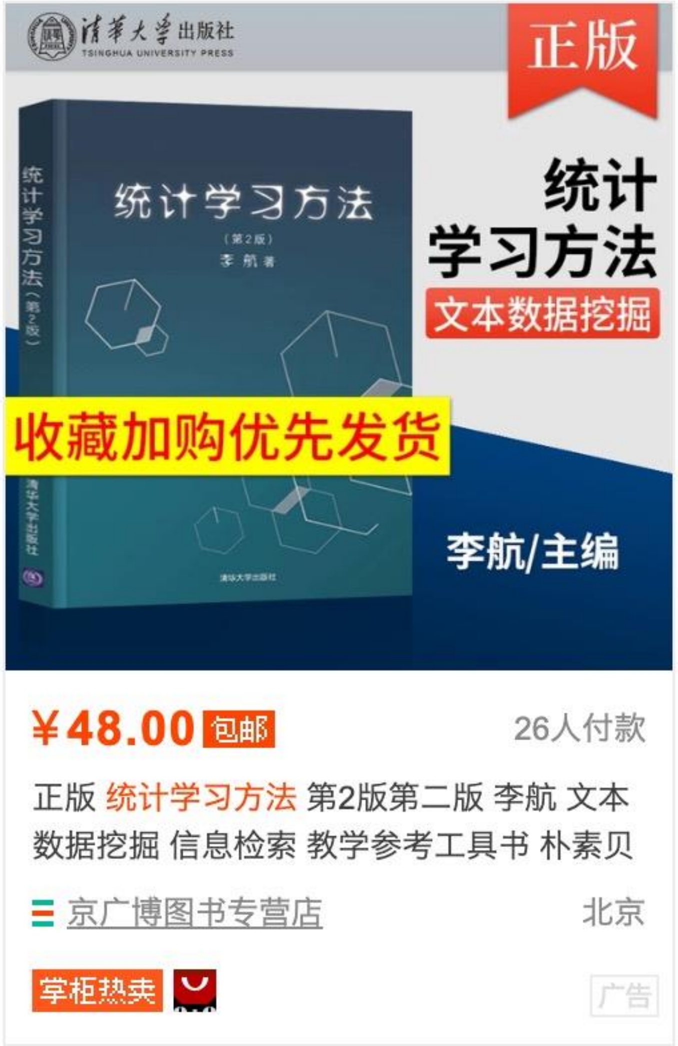
Course Objective

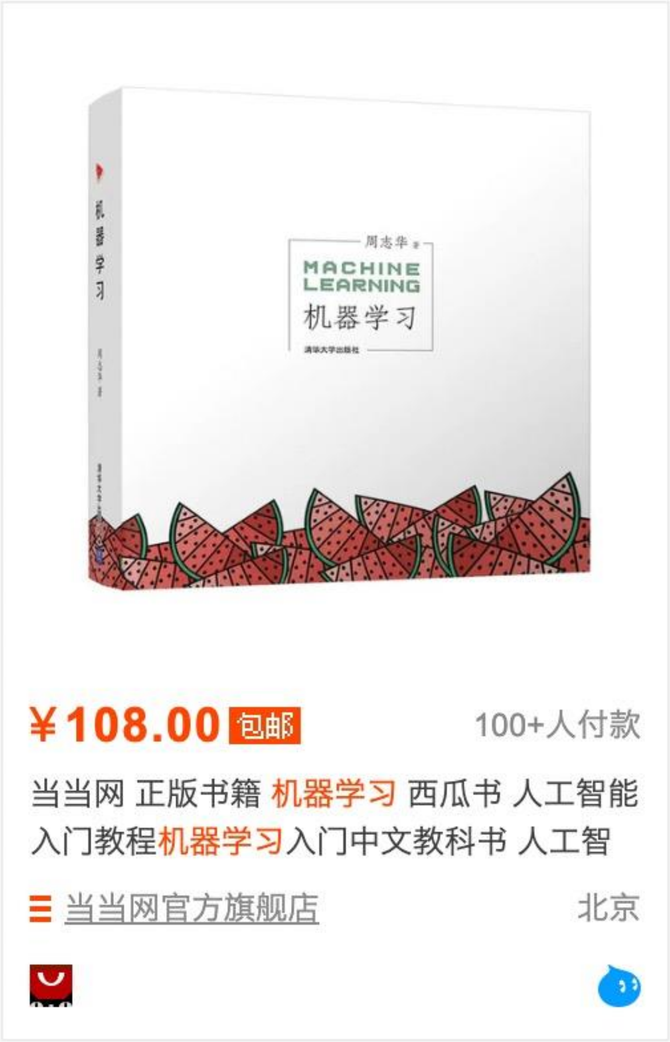
- The course gives an introduction to machine learning and its applications.
- The course aims to provide the students the basic ideas and intuition about machine learning algorithms as well as its implementation and application.
- Besides, there will be Python Workshops on various basic Machine Learning and AI Technologies/applications, such as Pytorch, Various Supervised, Unsupervised, and Reinforcement Learning technologies, Artificial Neural Networks, and Deep Networks.

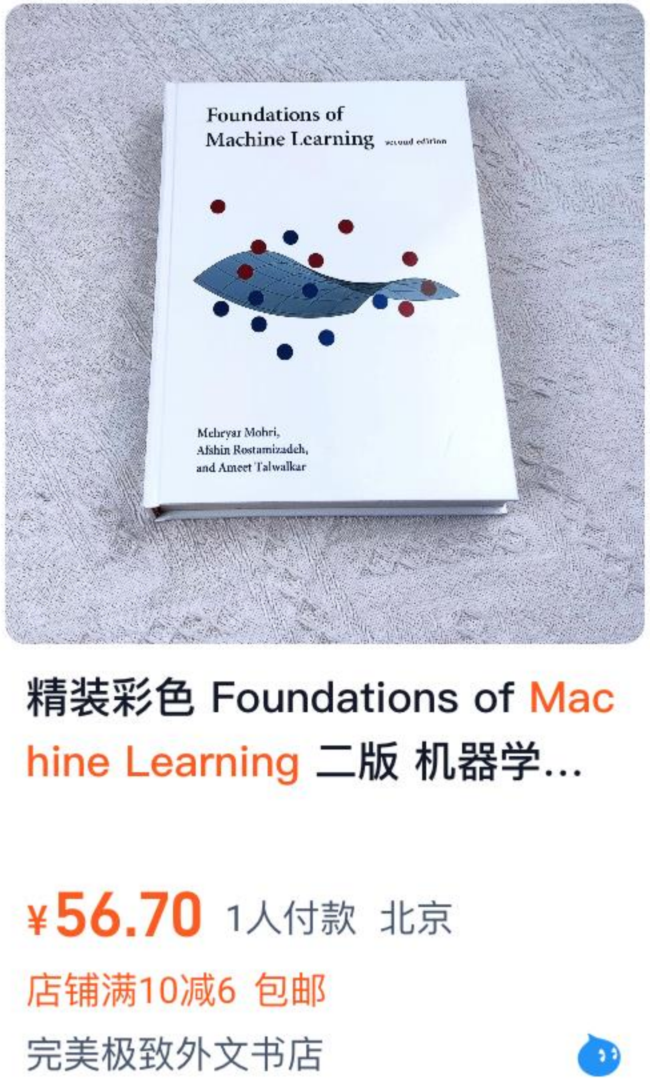
Overview of ML Family



Reference textbook (Chinese & English)







Recommended teaching materials

- 阿斯顿·张等,动手学深度学习, ISBN: 9787115505835.
  - Dive into Deep Learning, https://d2l.ai/
- Bishop, C.M. (2006). Pattern recognition and Machine Learning. Springer. ISBN 9780387310732.
- Goodfellow, Ian (2016). Deep Learning. MIT Press.
- 李航等, 统计学习方法(第二版), ISBN: 9787302517276.
- 周志华等, 机器学习, ISBN: 9787302423287.
- 龙曲良等，Tensorflow深度学习(深入理解人工智能算法设计).

Convex Optimization Textbooks：

- Wright, S., & Nocedal, J. (1999). Numerical optimization. Springer, 35(67-68), 7.
- Boyd, S., & Vandenberghe, L. (2004). Convex optimization. Cambridge University Press. (凸优化教材圣经)

Recommended Courses

- 斯坦福大学CS224n: Deep Learning for Natural Language Processing
  - https://web.stanford.edu/class/archive/cs/cs224n/cs224n.1194/
  - Chris Manning 主要讲解自然语言处理领域的各种深度学习模型
- 斯坦福大学CS231n: Convolutional Neural Networks for Visual Recognition
  - http://cs231n.stanford.edu/
  - Fei-Fei Li Andrej Karpathy 主要讲解CNN、RNN在图像领域的应用
- 加州大学伯克利分校 CS 294: Deep Reinforcement Learning
  - http://rail.eecs.berkeley.edu/deeprlcourse/

Recommended Materials

- 林轩田 “机器学习基石” “机器学习技法”
  - https://www.csie.ntu.edu.tw/~htlin/mooc/
- 李宏毅 “1天搞懂深度学习”
  - http://speech.ee.ntu.edu.tw/~tlkagk/slide/Tutorial_HYLee_Deep.pptx
- 李宏毅“机器学习2020”
  - https://www.bilibili.com/video/av94519857/

Prepare Mathematical Knowledge

- Linear Algebra: Matrix calculation
- Calculus: It doesn't need to be very deep, just the common differential integral
- Mathematical optimization: Gradient descent
- Probability theory: density function, expected variance, etc.
- Information theory: Cross entropy

Top Conferences

- NeurIPS、ICLR、ICML、AAAI、IJCAI
- ACL、EMNLP
- CVPR、ICCV
- …

中国计算机学会推荐国际学术会议

（人工智能）

一、A 类

序号	会议简称	会议全称	出版社	网址
1	AAAI	AAAI Conference on Artificial Intelligence	AAAI	http://dblp.uni-trier.de/db/conf/aaai/
2	NeurIPS	Annual Conference on Neural Information Processing Systems	MIT Press	http://dblp.uni-trier.de/db/conf/nips/
3	ACL	Annual Meeting of the Association for Computational Linguistics	ACL	http://dblp.uni-trier.de/db/conf/acl/
4	CVPR	IEEE Conference on Computer Vision and Pattern Recognition	IEEE	http://dblp.uni-trier.de/db/conf/cvpr/
5	ICCV	International Conference on Computer Vision	IEEE	http://dblp.uni-trier.de/db/conf/iccv/
6	ICML	International Conference on Machine Learning	ACM	http://dblp.uni-trier.de/db/conf/icml/
7	IJCAI	International Joint Conference on Artificial Intelligence	Morgan Kaufmann	http://dblp.uni-trier.de/db/conf/ijcai/

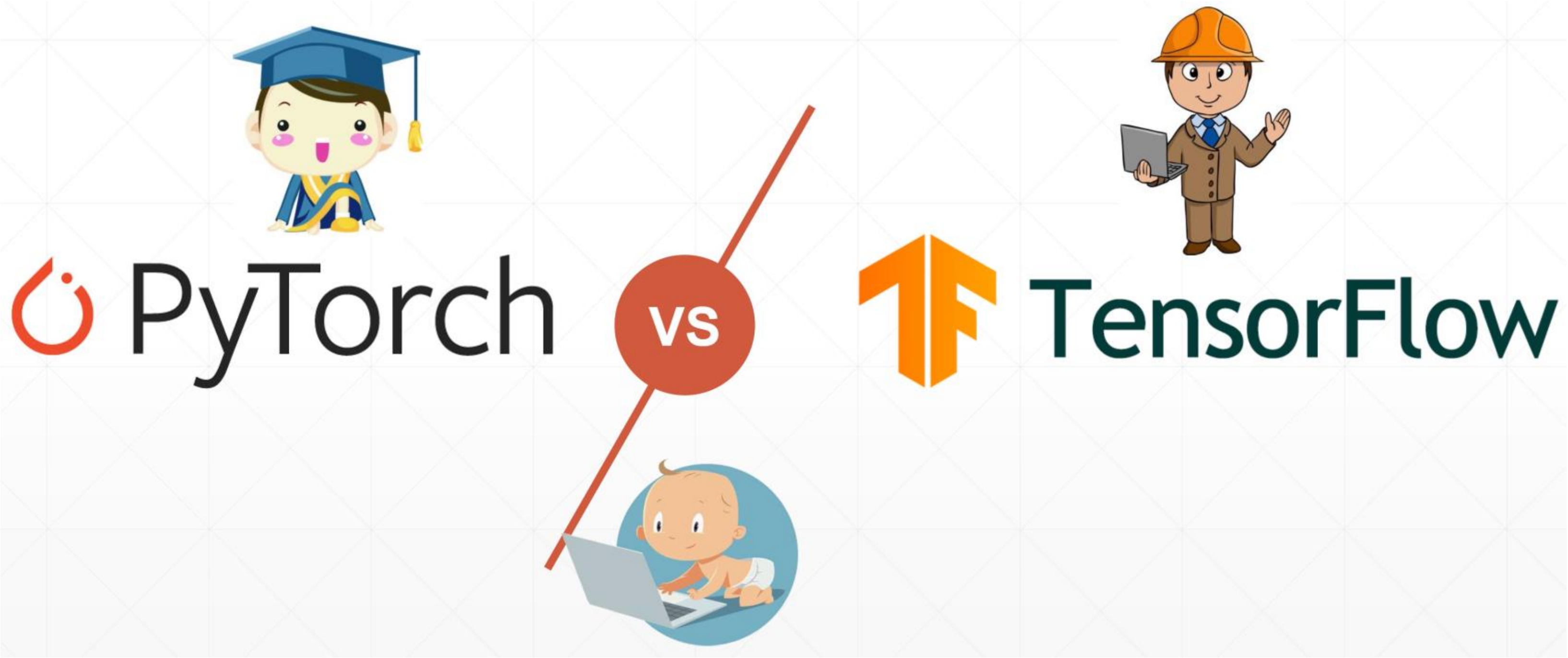
Machine Learning Framework

- Scikit-learn: https://scikit-learn.org/stable/
- The Python library, Scikit-Learn, is built on top of the matplotlib, NumPy, and SciPy libraries.
  - Simple, easy to use, and effective.
  - In rapid development, and constantly being improved.
  - Wide range of algorithms, including clustering, factor analysis, principal component analysis, and more.
  - Can extract data from images and text.
  - This library is especially suited for supervised learning, and not very suited to unsupervised learning applications like Deep Learning.

Deep Learning Framework

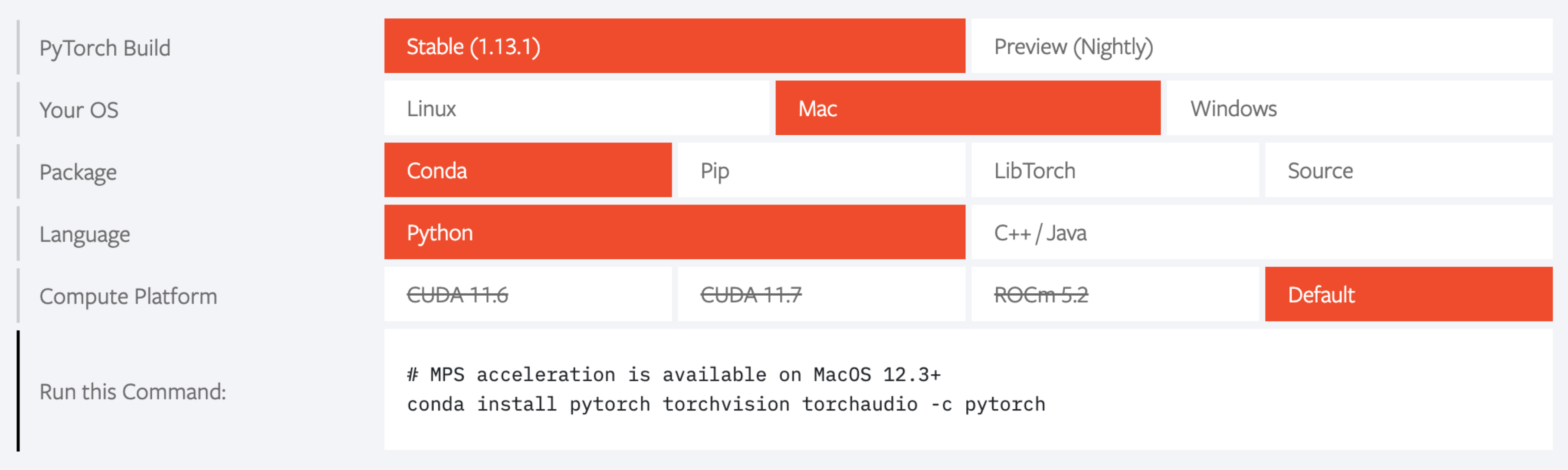
- Simple and fast prototype design
- Automatic gradient calculation
- Switching CPU and GPU
- Distributed

PaddlePaddle MindSpore



PyTorch Installation and Usage

- PyTorch website：https://pytorch.org/
- PyTorch document：https://pytorch.org/docs/stable/index.html
- Install: Python, Jypter notebook, Pytorch, etc.
- Install tutorial：https://www.bilibili.com/video/BV1Jg411c72R/?spm_id_from=333.337.search-card.all.click&vd_source=1c8c97959927ae70af2fd929f2a41663



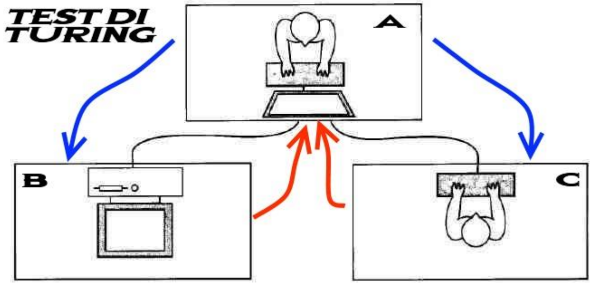
1.1 Artificial Intelligence

Turing Test

"A person has a series of questions and answers with each other in a special way without touching each other. If for a long time, he can't judge whether the other party is a person or a computer according to these questions, then he can think that this computer is intelligent. "

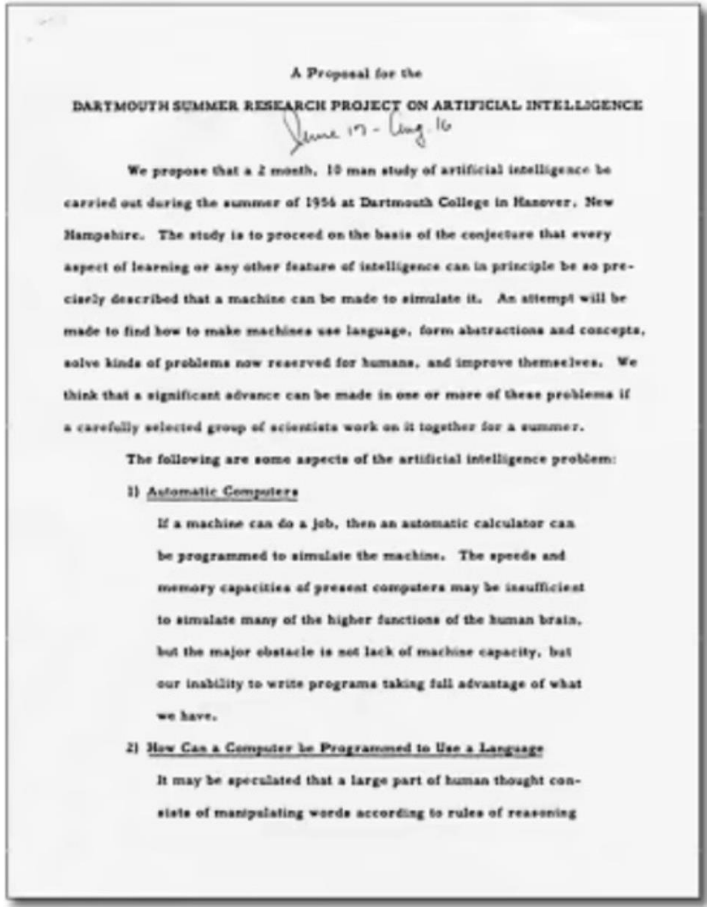
---Alan Turing [1950]
《Computing Machinery and Intelligence》

Alan Turing



Artificial Intelligence (AI)

- The birth of AI: The birth of AI has a clear landmark event: the 1956 Dartmouth conference. At this meeting, "artificial intelligence" was proposed as the name of this research field.



A Proposal for the

DARTMOUTH SUMMER RESEARCH PROJECT ON ARTIFICIAL INTELLIGENCE

June 17 - Aug 16

We propose that a 2 month, 10 man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it. An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. We think that a significant advance can be made in one or more of these problems if a carefully selected group of scientists work on it together for a summer.

The following are some aspects of the artificial intelligence problem:

1) Automatic Computers

If a machine can do a job, then an automatic calculator can be programmed to simulate the machine. The speeds and memory capacities of present computers may be insufficient to simulate many of the higher functions of the human brain, but the major obstacle is not lack of machine capacity, but our inability to write programs taking full advantage of what we have.

2) How Can a Computer be Programmed to Use a Language

It may be speculated that a large part of human thought consists of manipulating words according to rules of reasoning

Artificial Intelligence

- Artificial intelligence (AI) is to make machines have human intelligence.
- Computer Control + Intelligent Behavior

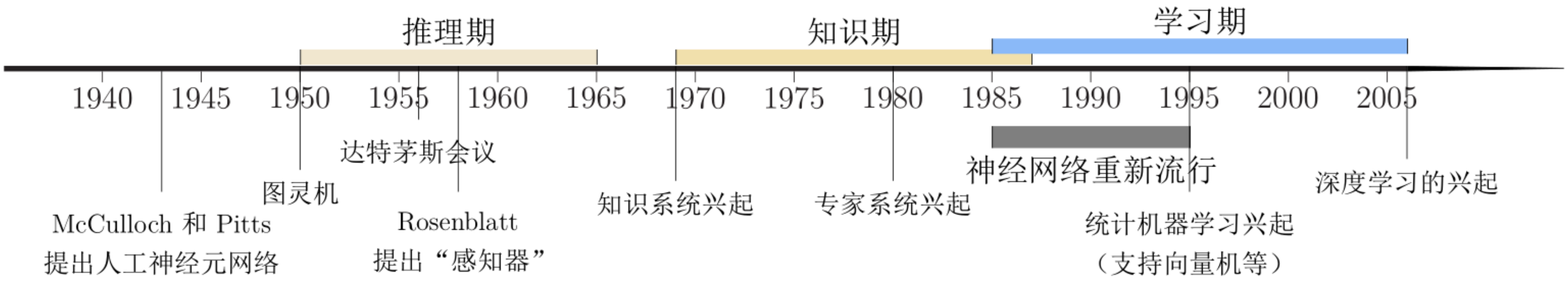
Artificial intelligence is to make the behavior of machines look like the intelligent behavior of people.

John McCarthy（1927-2011）

Research Areas in AI

- Turing test is an essential factor to promote AI from philosophical discussion to scientific research, and has guided many research directions of AI.
  - To make computer pass Turing test, computer must understand language, learn, remember, reason, and make decisions.
- Let machines have human intelligence
  - Machine Perception (CV, Speech info. Processing, Pattern Recog.)
  - Learning (ML, Reinforcement Learning (RL))
  - Language (Natural Language Processing (NLP))
  - Memory (Knowledge Representation）
  - Decision-making (Planning, Data Mining)

Development History

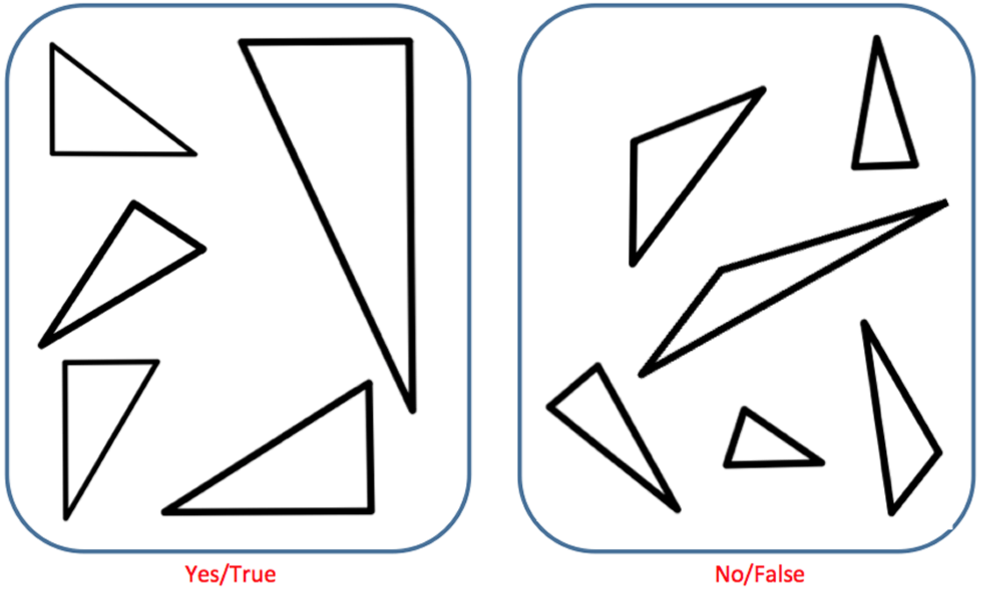


1.2 How to develop AI system

How to develop an AI system?

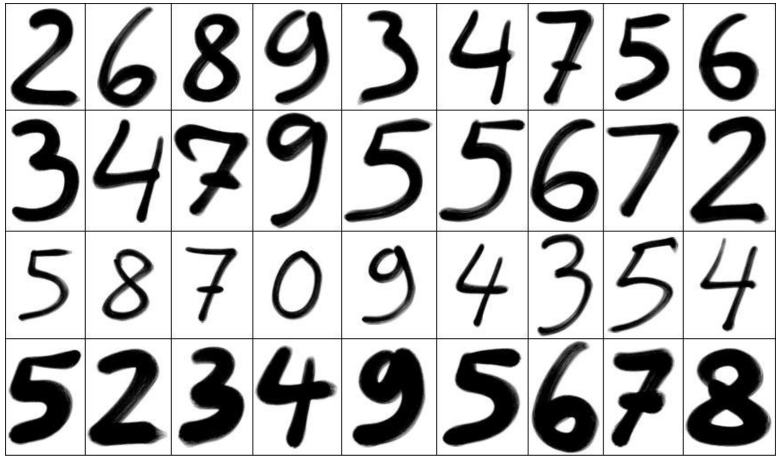
- Expert knowledge (manual rules)

What's My Rule?



What’s the Rule?

Machine Learning!



Machine Learning ≈ Construct a mapping function (rule)

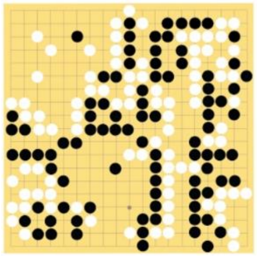
- Speech recognition

$f($  $)=$ “你好”

- Image recognition

$f($  $)=$ “9”

- 围棋

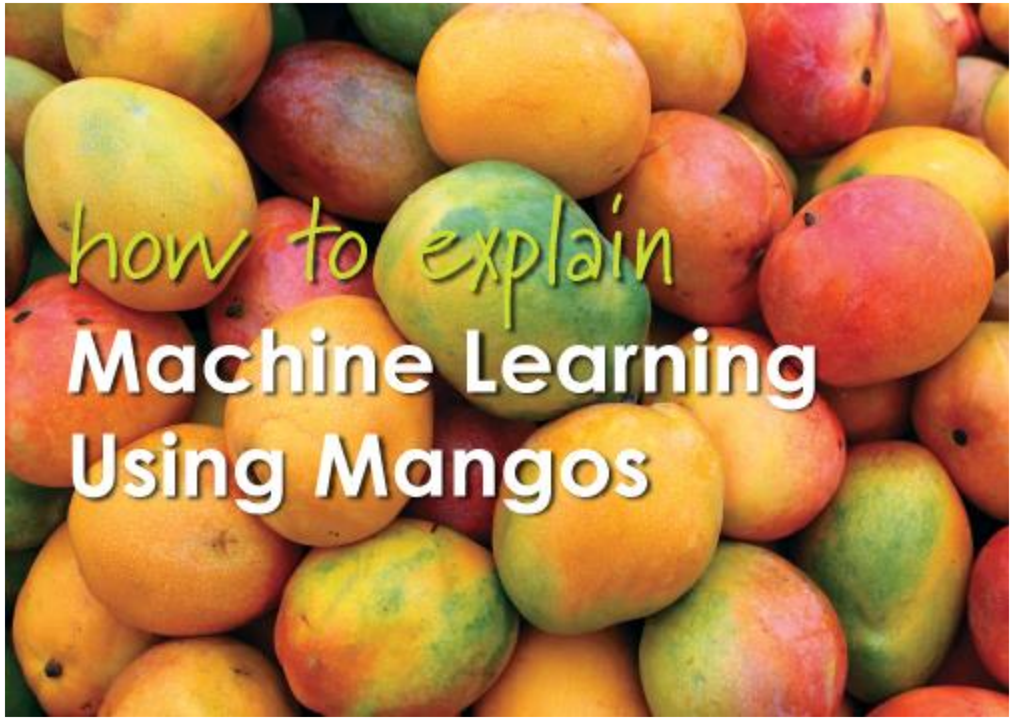
$f($  $)=$ “6-5” (落子位置)

- Machine Translation

$f($ “你好！” $)=$ “Hello!”

Mango ML

How to judge whether mango is sweet?



https://www.quora.com/How-do-you-explain-Machine-Learning-and-Data-Mining-to-non-Computer-Science-people

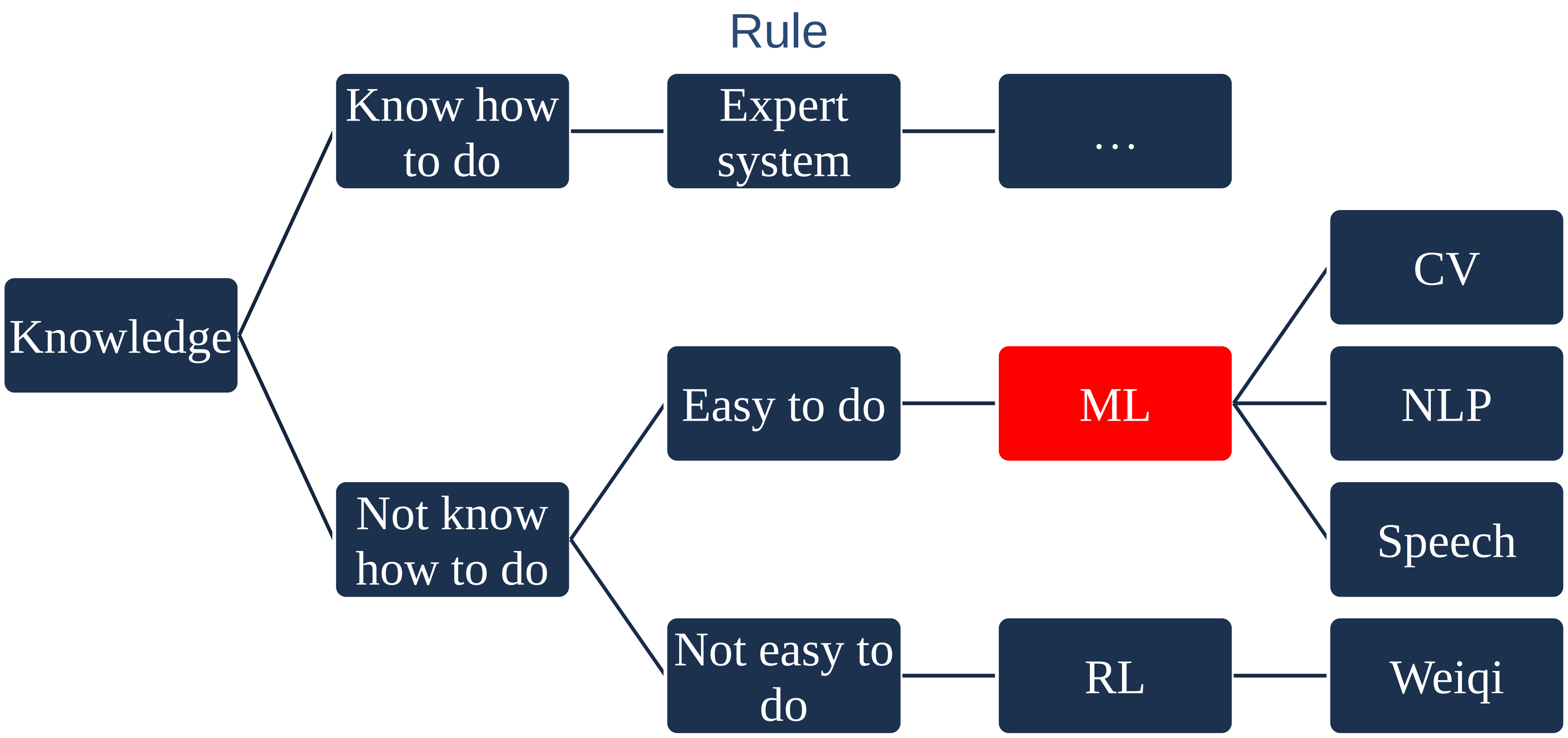
Mango ML

- Prepare data for training
  - Randomly selected mango samples from the market (training data), list all the characteristics of each mango:
    - Such as color, size, shape, origin, brand, and price
  - Mango quality (output variable):
    - Sweet, juicy, and mature
- Learning
  - Design a learning algorithm to learn the correlation model between mango features and output variables.

Mango ML

- Testing
  - The next time you buy mango from the market, you can use the model calculated above to predict the quality of mango according to its characteristics (test data). (Accuracy)

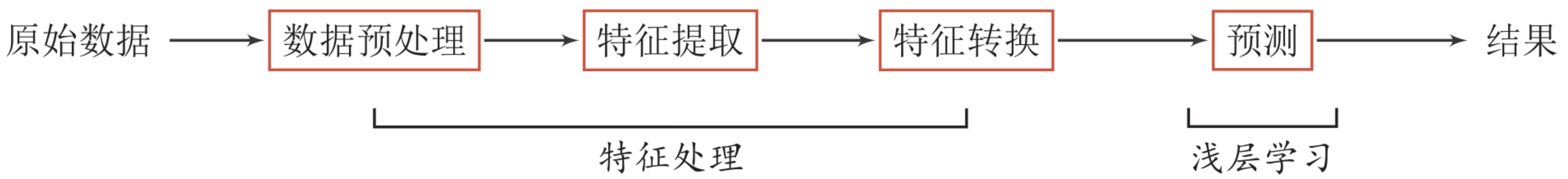
How to develop an AI system?



1.3 Representation Learning

Machine Learning (ML)

- When we use ML to solve some pattern recognition tasks, the general process includes the following steps:



特征工程（Feature Engineering）

- Shallow Learning: It does not involve feature learning, and its features are mainly extracted by manual experience or feature conversion method.

Semantic Gap: One of the Challenges of AI

- Low-level characteristics VS High-level Semantics
  - People's understanding of texts and images cannot be directly obtained from underlying features of character strings or images.

床前明月光，

疑是地上霜。

举头望明月，

低头思故乡。

What is a good representation?

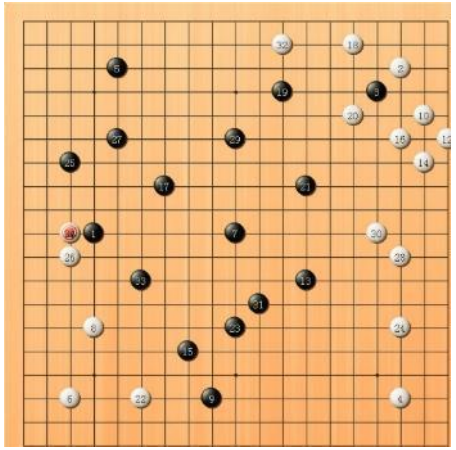
- "Good representation" is a very subjective concept and there is no clear standard.
- But a good representation has the following advantages:
  - Have strong presentation ability
  - Make the follow-up learning tasks simple
  - be general and independent of tasks or fields
- Data representation is the core problem of ML
  - Feature engineering: the need for human intelligence

Representation Form

- **Local Representation**
  - Discrete representation
  - Symbolic representation
  - One-Hot vector

	Local	Distributed
A	[1 0 0 0]	[0.25 0.5]
B	[0 1 0 0]	[0.2 0.9]
C	[0 0 1 0]	[0.8 0.2]
D	[0 0 0 1]	[0.9 0.1]

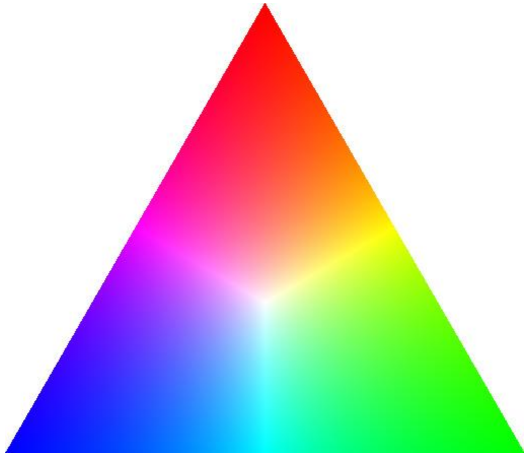
- **Distributed Representation**
  - Compressed, low-dimension, dense vector
  - Use O(N) parameters to show O(2^k) interval
    - k is a non-zero parameter, $k < N$



Distributed Representation

An example in life: Color

颜色	局部表示	分布式表示
琥珀色	$[1,\ 0,\ 0,\ 0]^{\mathrm{T}}$	$[1.00,\ 0.75,\ 0.00]^{\mathrm{T}}$
天蓝色	$[0,\ 1,\ 0,\ 0]^{\mathrm{T}}$	$[0.00,\ 0.5,\ 1.00]^{\mathrm{T}}$
中国红	$[0,\ 0,\ 1,\ 0]^{\mathrm{T}}$	$[0.67,\ 0.22,\ 0.12]^{\mathrm{T}}$
咖啡色	$[0,\ 0,\ 0,\ 1]^{\mathrm{T}}$	$[0.44,\ 0.31\ 0.22]^{\mathrm{T}}$



Semantic Representation

- How to express semantics in a computer?

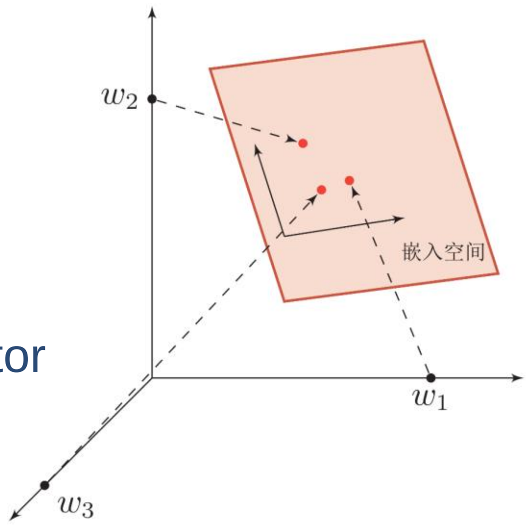
Local Representation → Distributed Representation

Knowledge base, rules

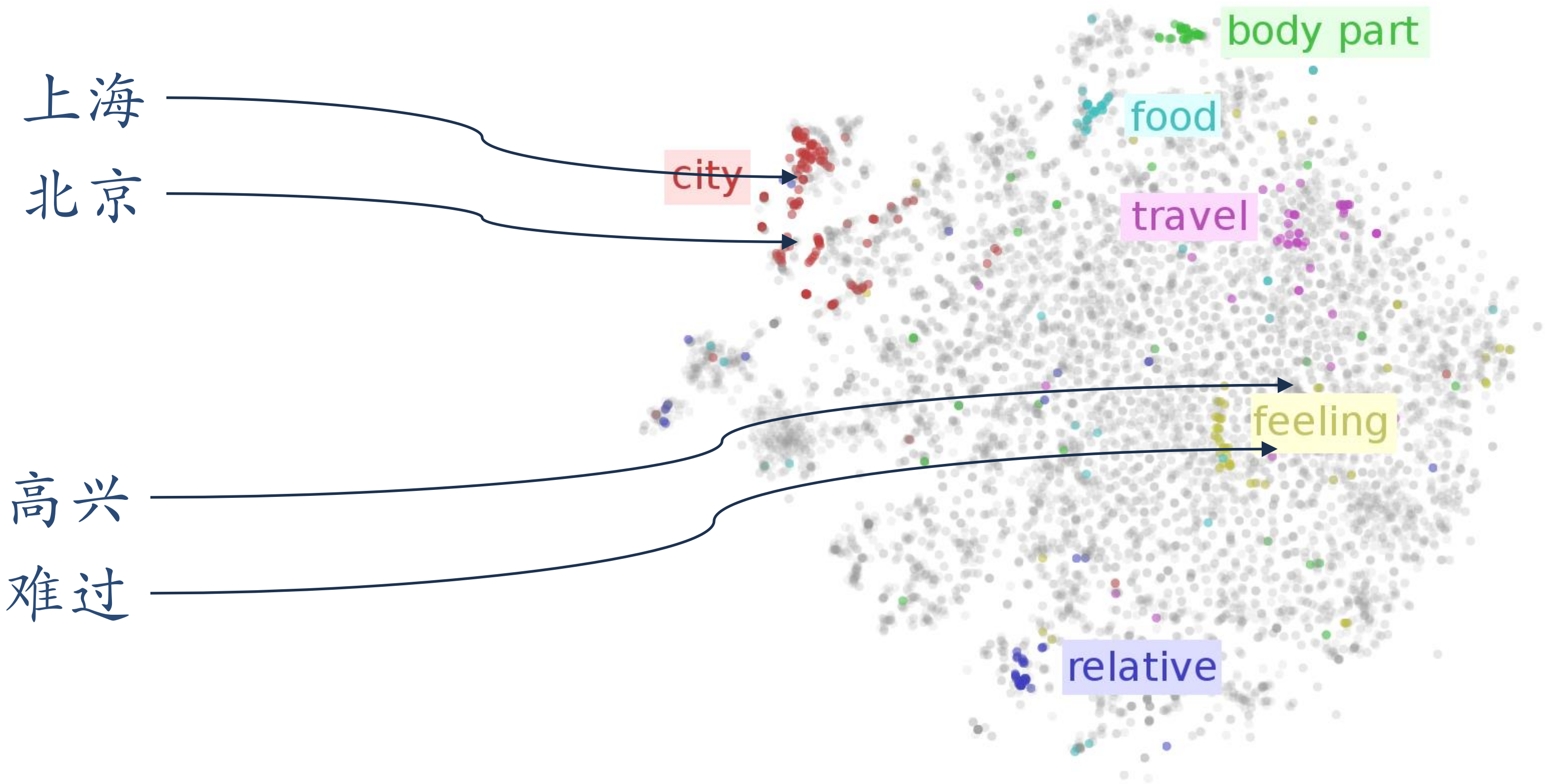
Embedding: compressed, low-dimensional, dense vector

Embedding

One-Hot vector



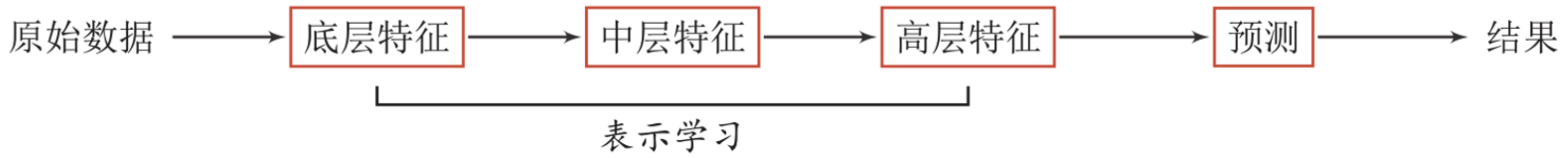
Word Embeddings



https://indico.io/blog/visualizing-with-t-sne/

Representation Learning

- Representation learning: how to automatically learn good representations from data
- By constructing a model which can automatically learn good feature representation (from bottom features to middle features and then to high features), thus ultimately improving the accuracy of prediction or recognition.



Bengio, Yoshua, Aaron Courville, and Pascal Vincent. "Representation learning: A review and new perspectives." IEEE transactions on pattern analysis and machine intelligence 35.8 (2013): 1798-1828.

Traditional Feature Extraction

- Feature extraction
  - Linear projection (Subspace)
    - PCA、LDA
  - Nonlinear embedding
    - LLE、Isomap、Spectral methods
  - Anto-encoder
- Feature Extraction VS Representation Learning
  - Feature extraction: removing useless features based on tasks or prior pairs.
  - Representation learning: learning high-level semantic features through depth model.

Difficulties: there is no clear goal!

1.4 Machine Learning

Machine Learning ≈ Construct a mapping function (rule)

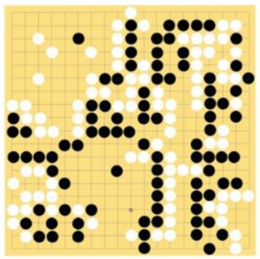
- Speech recognition

$f($  $)=$ “你好”

- Image recognition

$f($  $)=$ “9”

- 围棋

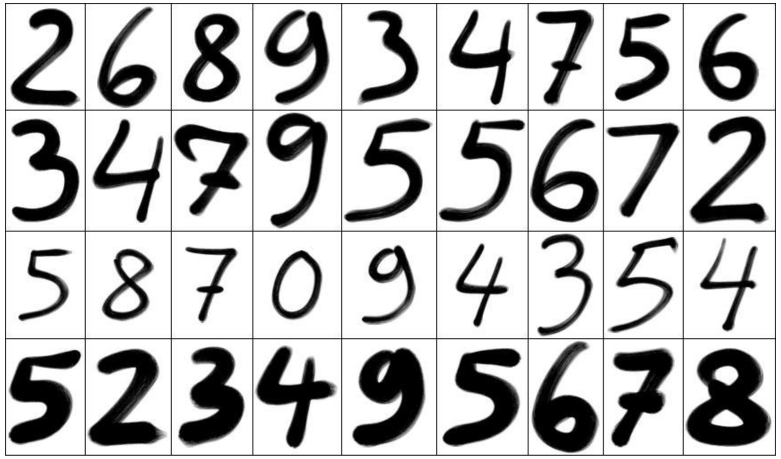
$f($  $)=$ “6-5” （落子位置）

- Machine Translation

$f($ “你好！” $)=$ “Hello!”

Why do we need ML?

- The problems in the real world are more complicated. Difficult to achieve it manually through rules.
- Map function, how?
- Learn rules from a large number of data



Definition of ML

Machine learning (ML) is the study of computer algorithms that improve automatically through experience and by the use of data.

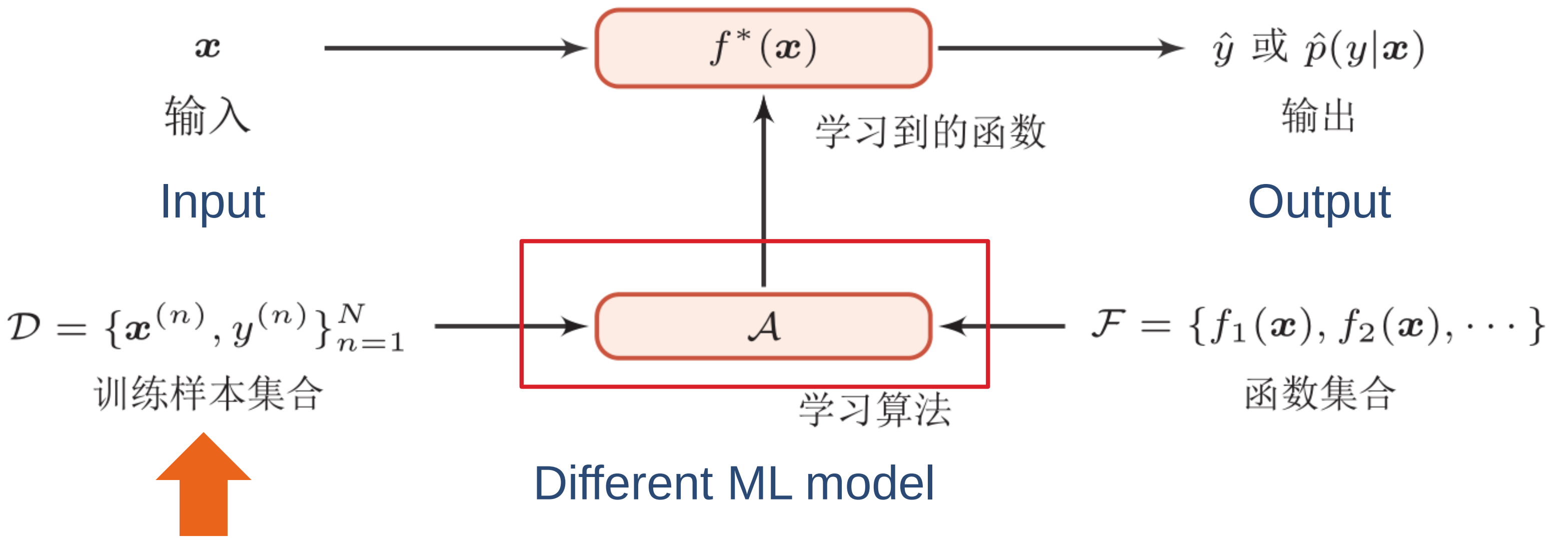
https://en.wikipedia.org/wiki/Machine_learning

WIKIPEDIA
The Free Encyclopedia

机器学习（Machine Learning，ML）是指从有限的观测数据中学习（或“猜测”）出具有一般性的规律，并利用这些规律对未知数据进行预测的方法.

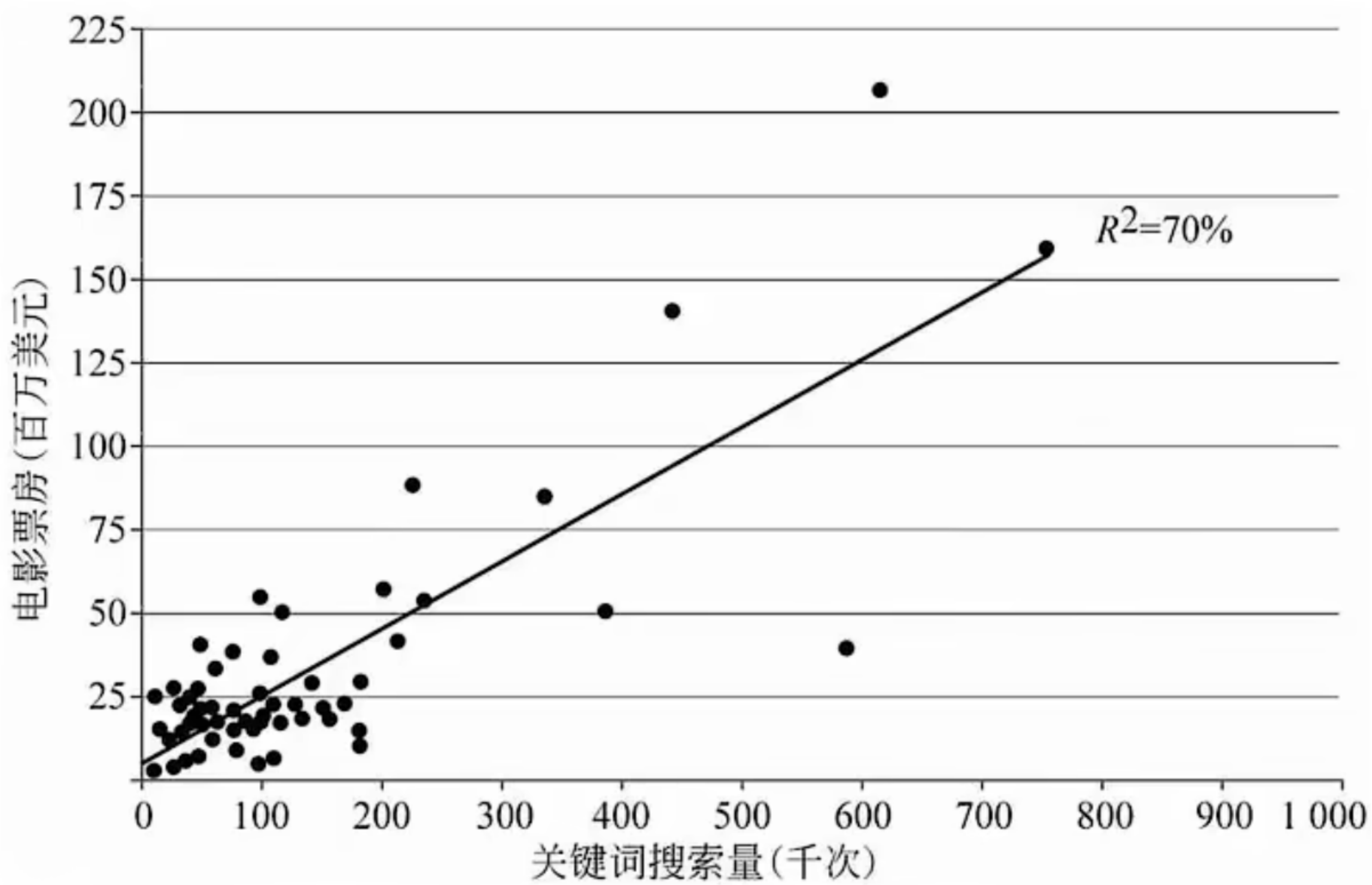
Definition of ML

- ML: through algorithms, machines can learn rules from a large number of data and make decisions on new sample
- Rules: decision (prediction) function



Independent and Identical distribution p(x,y)

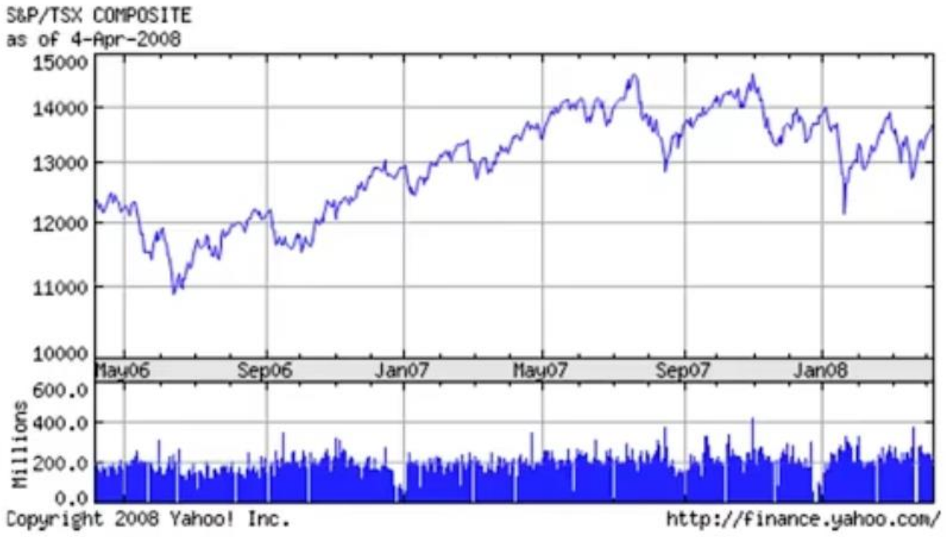
Example: Movie sale prediction



评分
导演影响力
演员影响力
电影档期
电影投资方
电影续集
电影制作技术
R–squared
Adjusted R–squared
F–statistic
Prob(F–statistic)

回归（Regression）问题

Example: Stock Price Prediction



公司盈利
负债
现金流
销售额
利润率

⇨ 股价

回归（Regression）问题

Example: House Price Prediction

Size (feet2)	Number of bedrooms	Number of floors	Age of home (years)	Price ($1000)
2104	5	1	45	460
1416	3	2	40	232
1534	3	2	30	315
852	2	1	36	178
…	…	…	…	…

回归（Regression）问题

Example: Handwritten Digit Recognition

518 IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, VOL. 24, NO. 24, APRIL 2002

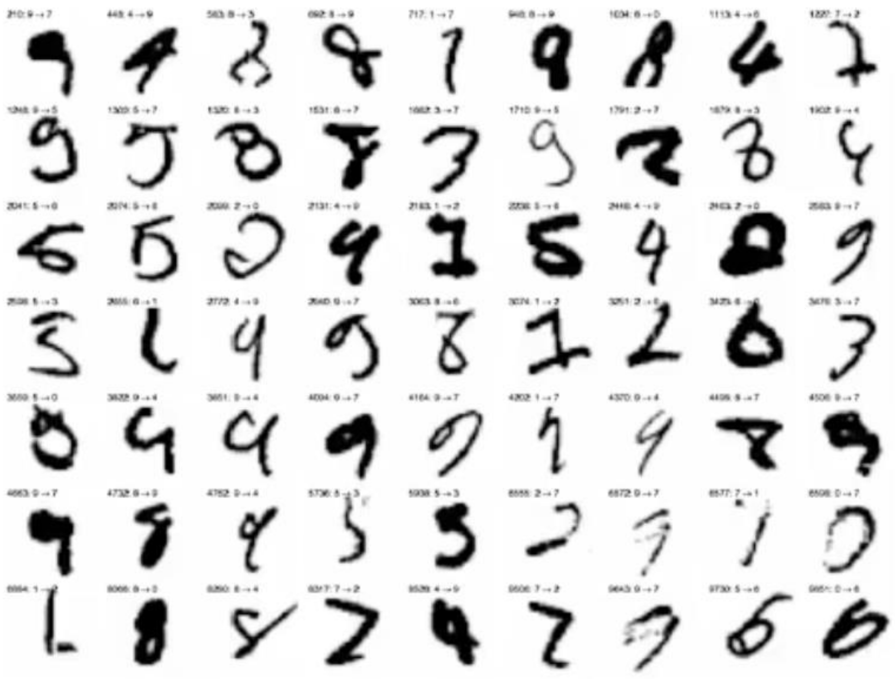
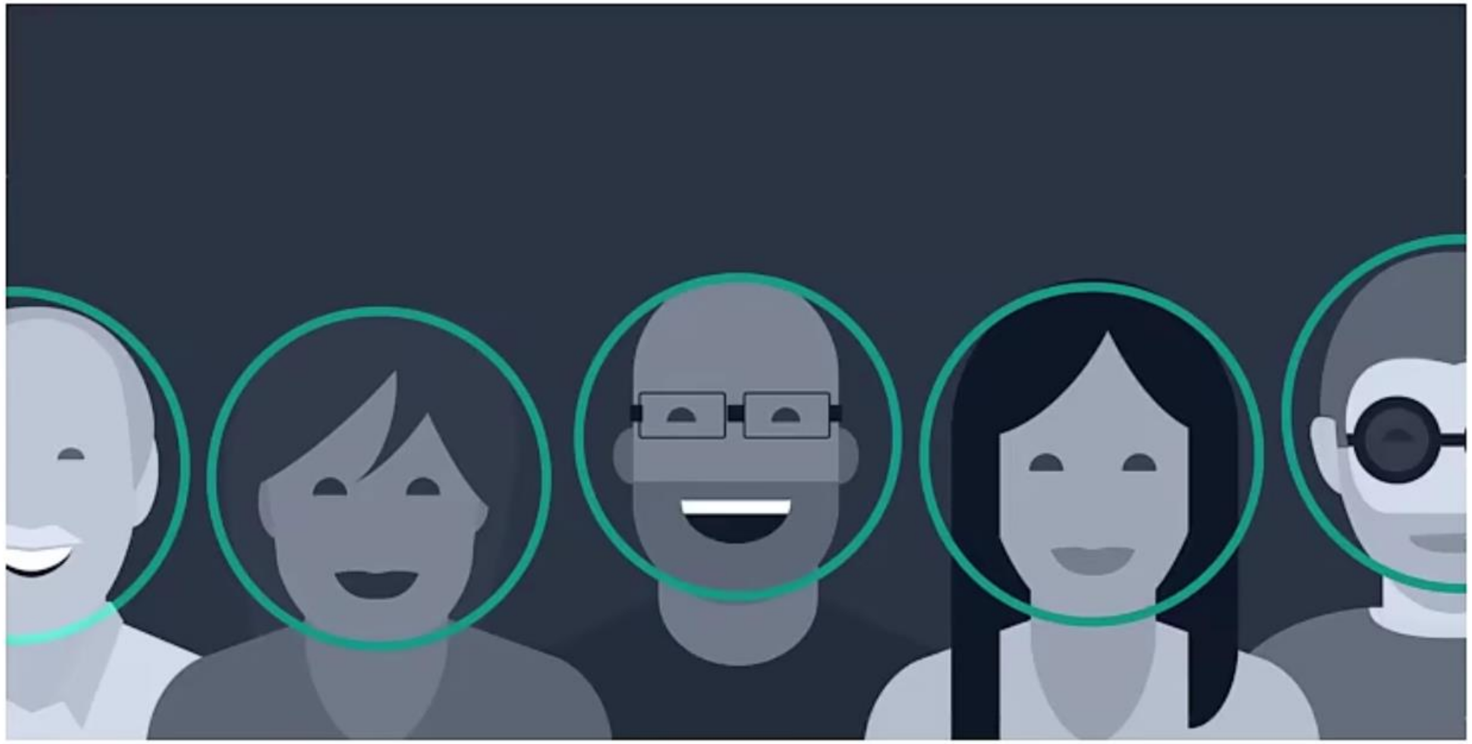


Fig. 8. All of the misclassified MNIST test digits using our method (63 out of 10,000). The text above each digit indicates the example number followed by the true label and the assigned label.

Belongie et al. PAMI 2002

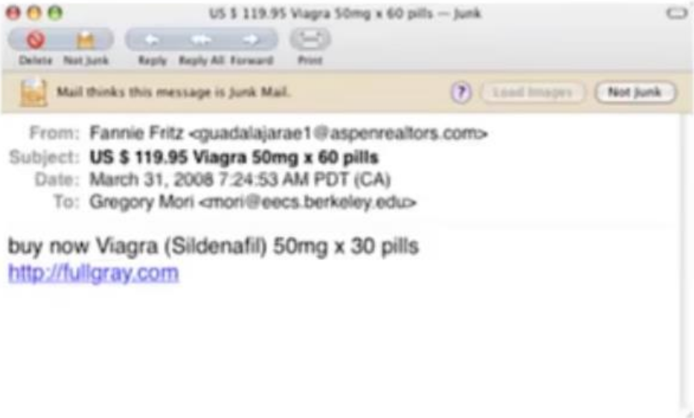
分类（Classification）问题

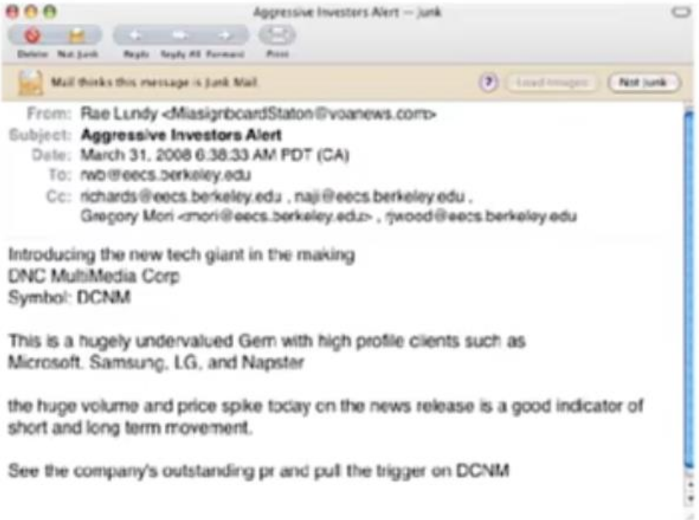
Example: Face Detection



分类（Classification）问题

Example: Spam Detection

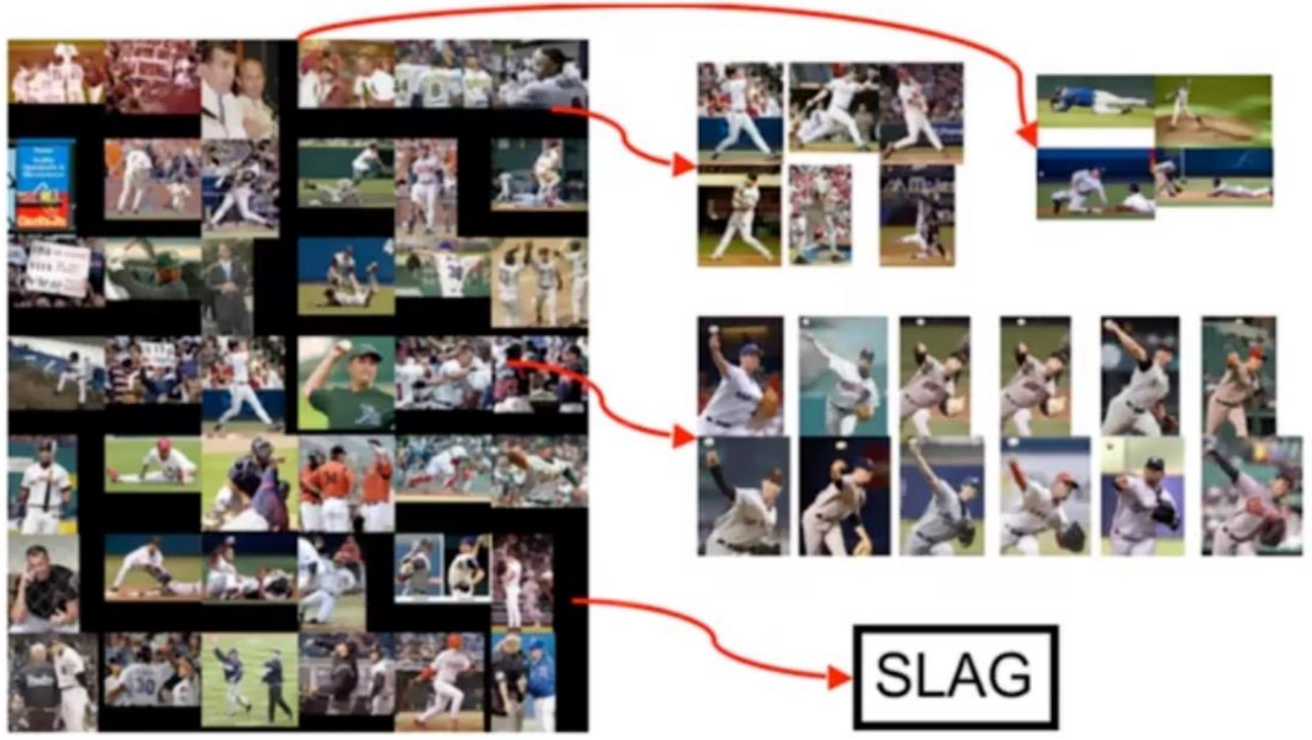




- Classification problem
- $t_i \in \{0, 1\}$, non-spam, spam
- x_i counts of words, e.g. `Viagra`, `stock`, `outperform`, `multi-bagger`

分类（Classification）问题

Example: Image Clustering



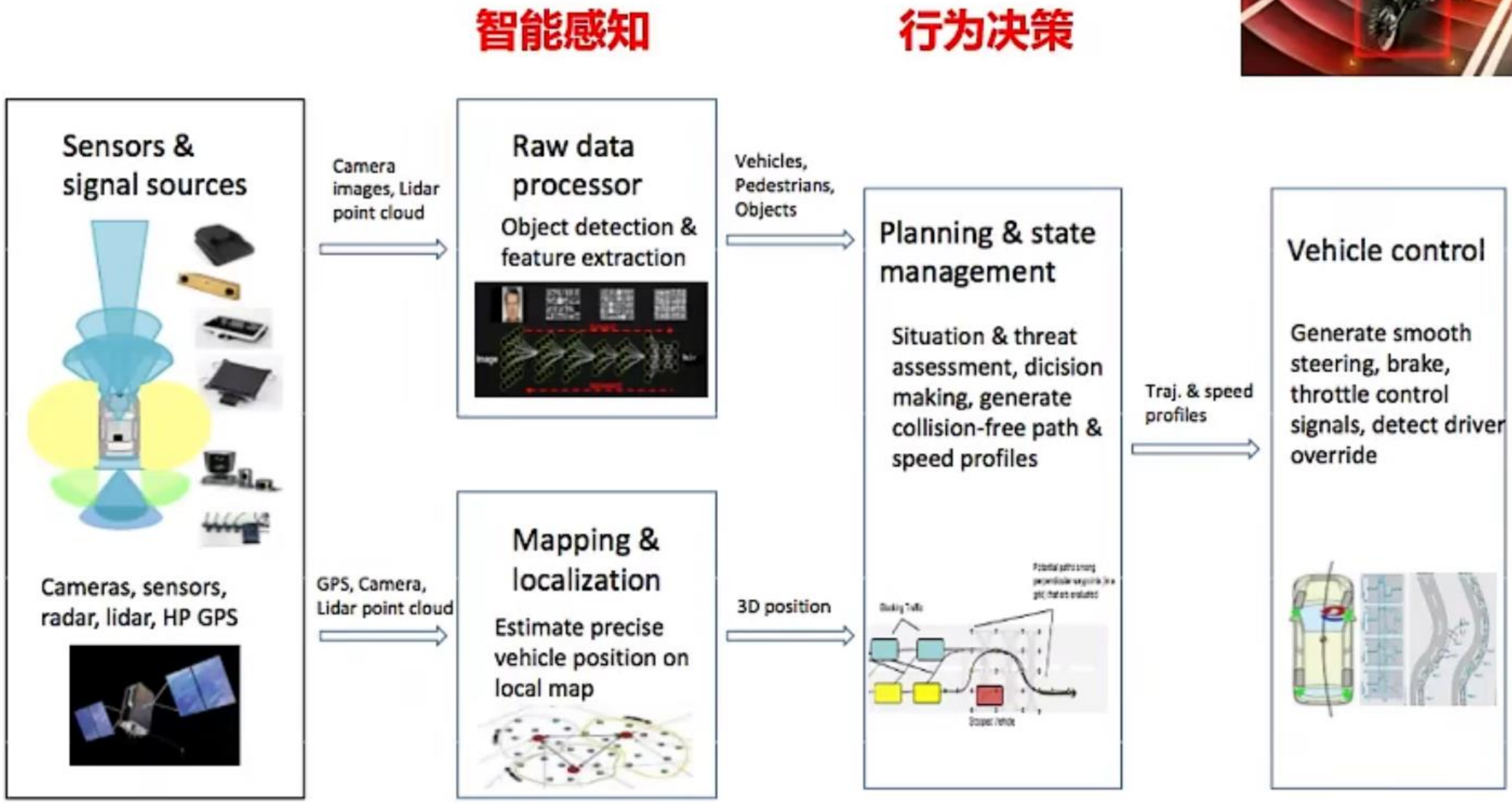
Wang et al., CVPR 2006

- Only $\boldsymbol{x}_i$ is defined: unsupervised learning
- E.g. $\boldsymbol{x}_i$ describes image, find groups of similar images

Example: AlphaGo

强化学习-通过与环境进行交互来学习

Example: Auto-Driving



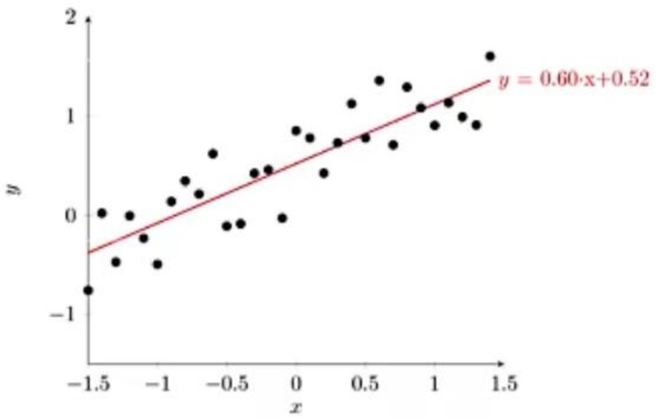
ML Problem Summarization

	Supervised Learning	**Unsupervised Learning**	**Reinforcement Learning**
Training sample	Training set $\{(\boldsymbol{x}_i, y_i)\}_{i=1}^N$	Training set $\{\boldsymbol{x}_i\}_{i=1}^N$	Trajectory of interaction between agent and environment τ and cumulative reward G_τ
Optimization goal	$y = f(\boldsymbol{x})$ or $p(y\|\boldsymbol{x})$	$p(\boldsymbol{x})$ or $p(\boldsymbol{x}\|\boldsymbol{z})$ with hidden variable $\boldsymbol{z}$	Expected cumulative reward $\mathbb{E}[G_\tau]$
Learning rule	Expected risk minimization or maximum likelihood estimation	maximum likelihood estimation or Minimum reconstruction error	Policy evaluation and Policy improvement

ML Problem Summarization

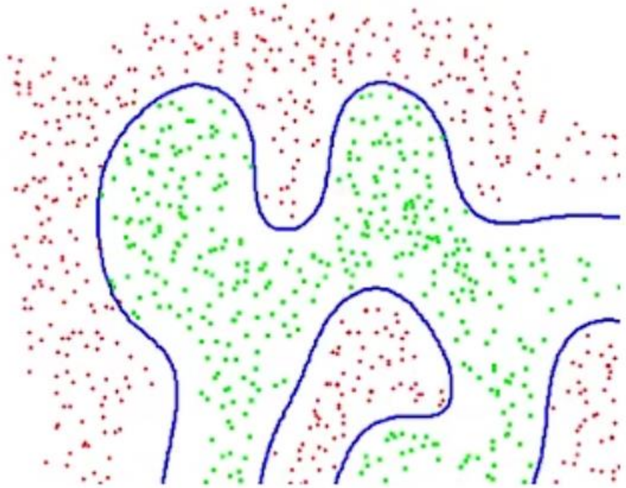
	监督学习	无监督学习	强化学习
训练样本	训练集 $\{(\mathbf{x}^{(n)}, y^{(n)})\}_{n=1}^{N}$	训练集 $\{\mathbf{x}^{n}\}_{n=1}^{N}$	智能体和环境交互的 轨迹 τ 和累积奖励 G_τ
优化目标	$y = f(\mathbf{x})$ 或 $p(y\|\mathbf{x})$	$p(\mathbf{x})$ 或带隐变量 $\mathbf{z}$ 的 $p(\mathbf{x}\|\mathbf{z})$	期望总回报 $\mathbb{E}_\tau[G_\tau]$
学习准则	期望风险最小化 最大似然估计	最大似然估计 最小重构错误	策略评估 策略改进

Typical Supervised Learning



回归

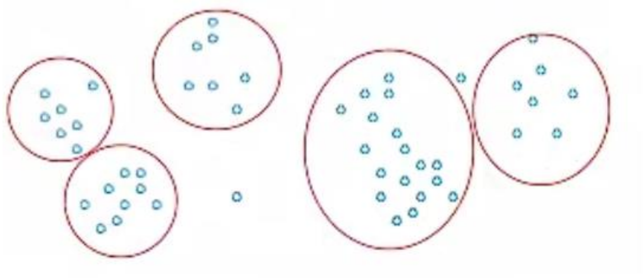
Regression



分类

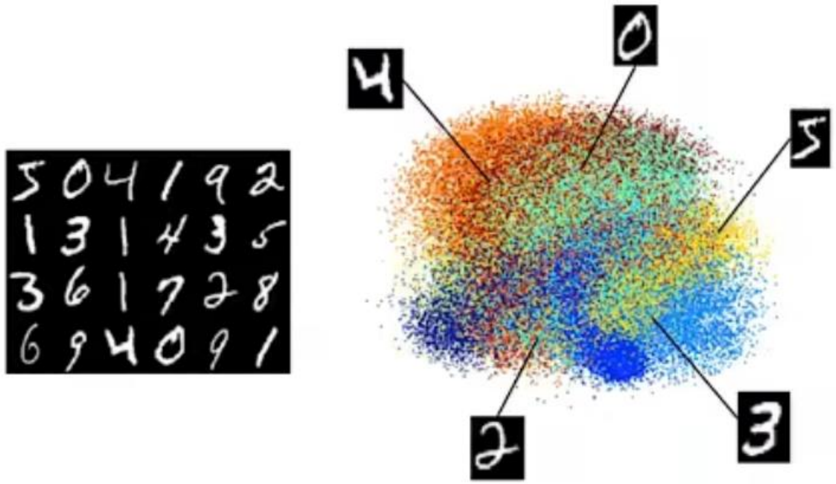
Classification

Typical Unsupervised Learning



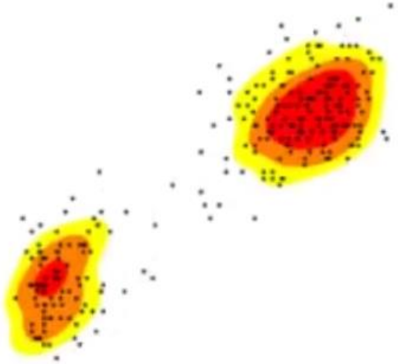
聚类

Clustering



降维

Dimension Reduction



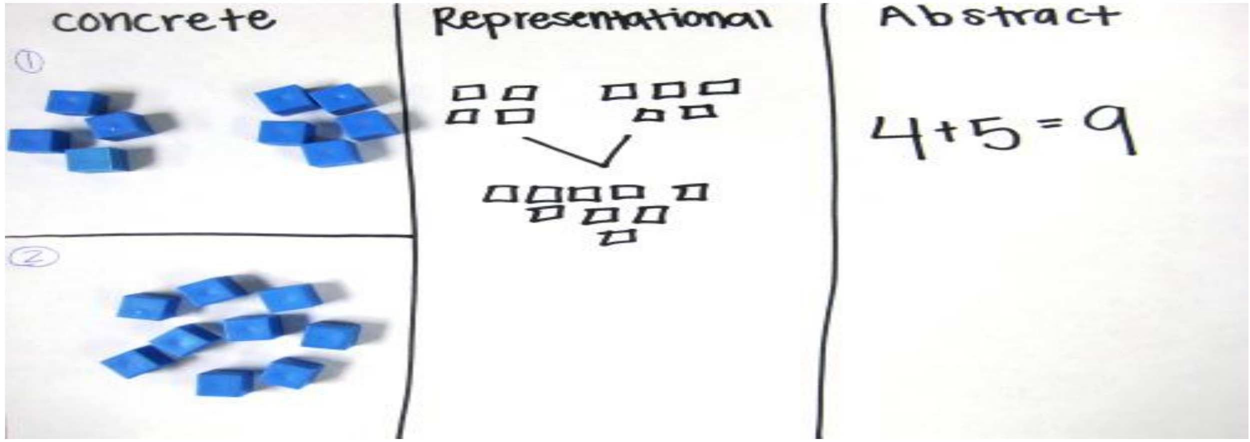
密度估计

Density Estimation

1.5 Deep Learning

Representation Learning v.s. Deep Learning

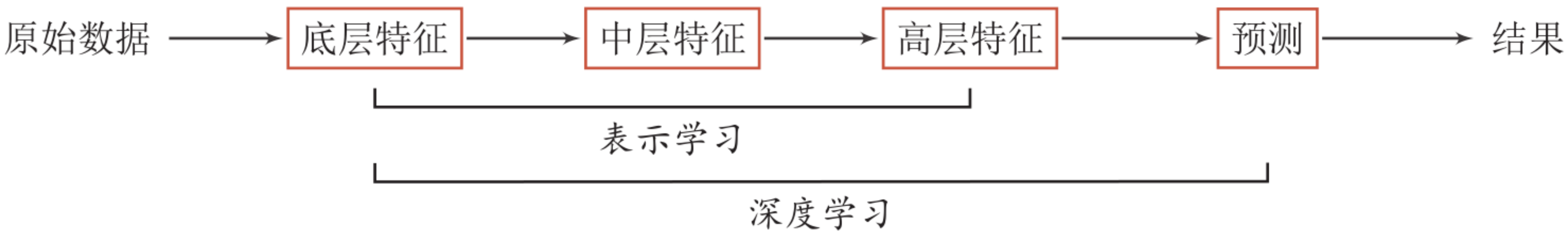
- A good representation learning strategy must have a certain depth
  - Feature reuse
    - Exponential representation ability
  - Abstract representation and invariance
    - Abstract representation requires multi-step construction



https://mathteachingstrategies.wordpress.com/2008/11/24/concrete-and-abstract-representations-using-mathematical-tools/

Deep Learning

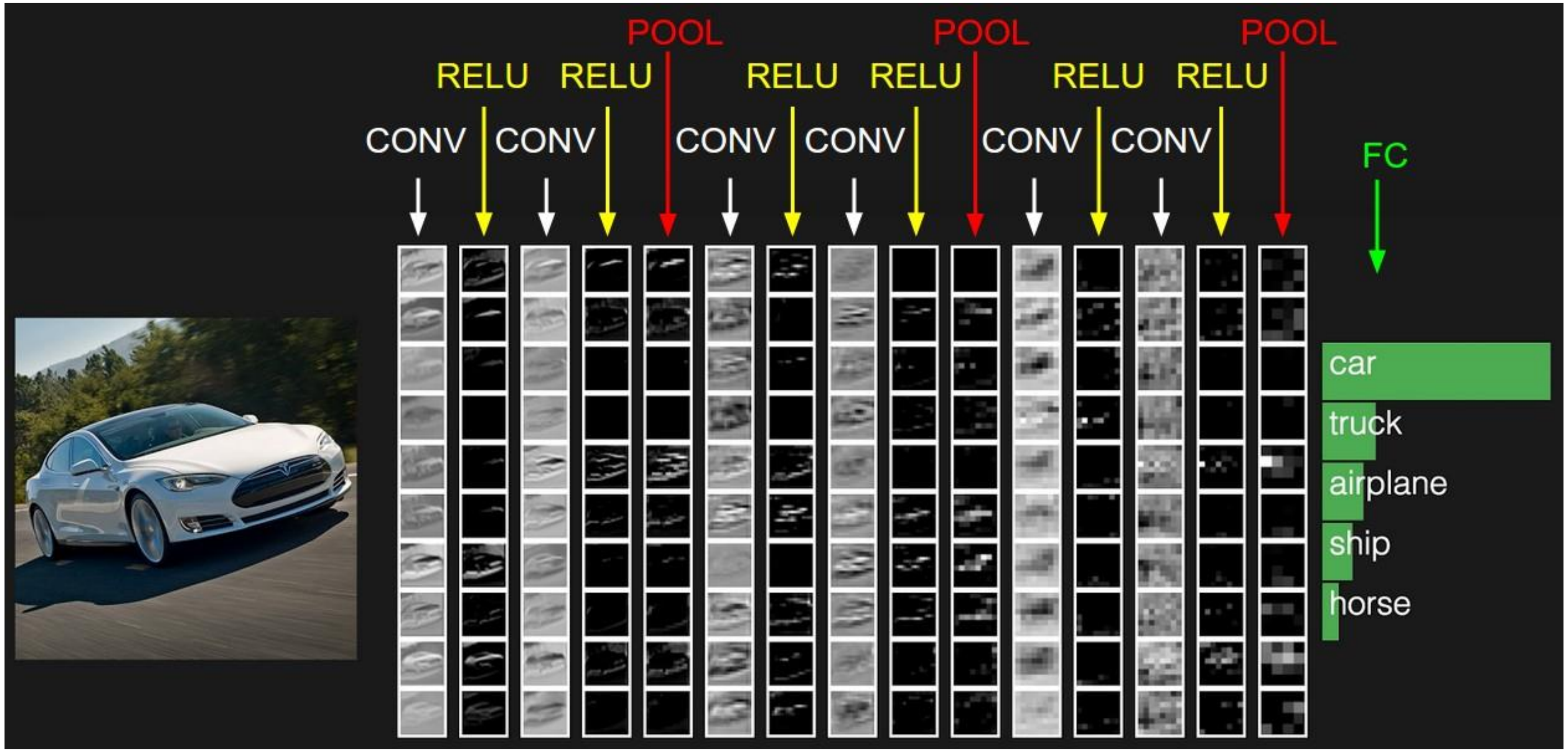
- Deep Learning = Representation Learning + Decision/Prediction Learning



Contribution distribution problem

贡献度分配问题

Representation Learning v.s. Deep Learning (End-to-End)



Mathematical Description of Deep Learning

$$y = f(x)$$

Shallow Learning

$$y = f^2(f^1(x))$$

$$\cdots$$

$$y = f^5(f^4(f^3(f^2(f^1(x)))))$$

Deep Learning

$f^{l}(x)$ is non-linear function, and not necessarily continuous

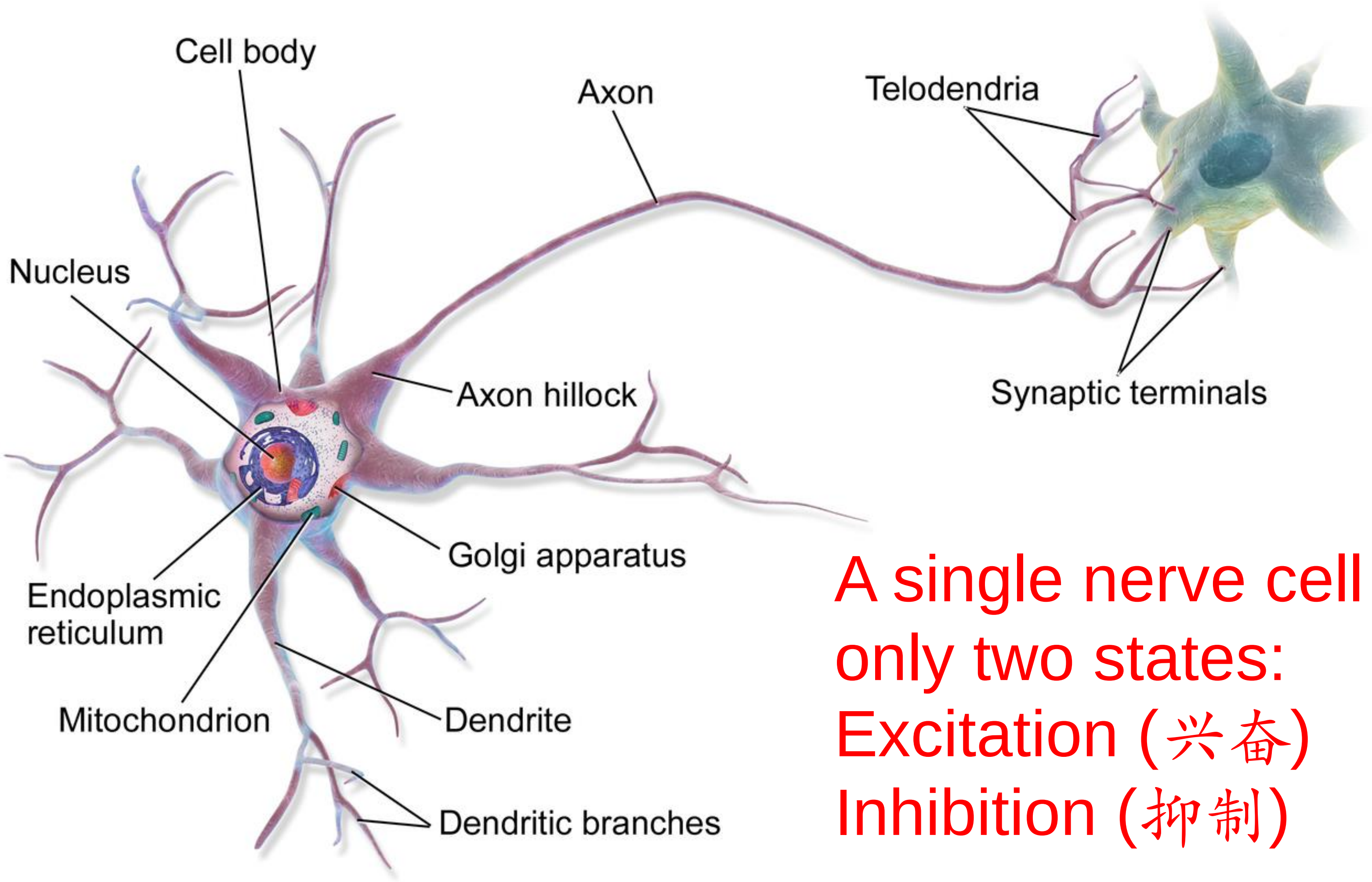
When $f^l(x)$ is continuous, such as $f^l(x) = \sigma\left(W^{(l)} f^{l-1}(x)\right)$, this composite function is called "Neural Network".

1.6 Neural Networks

Biological Neuron

video: structure of brain

The human brain has 86 billion neurons.



A single nerve cell has only two states:
Excitation (兴奋)
Inhibition (抑制)

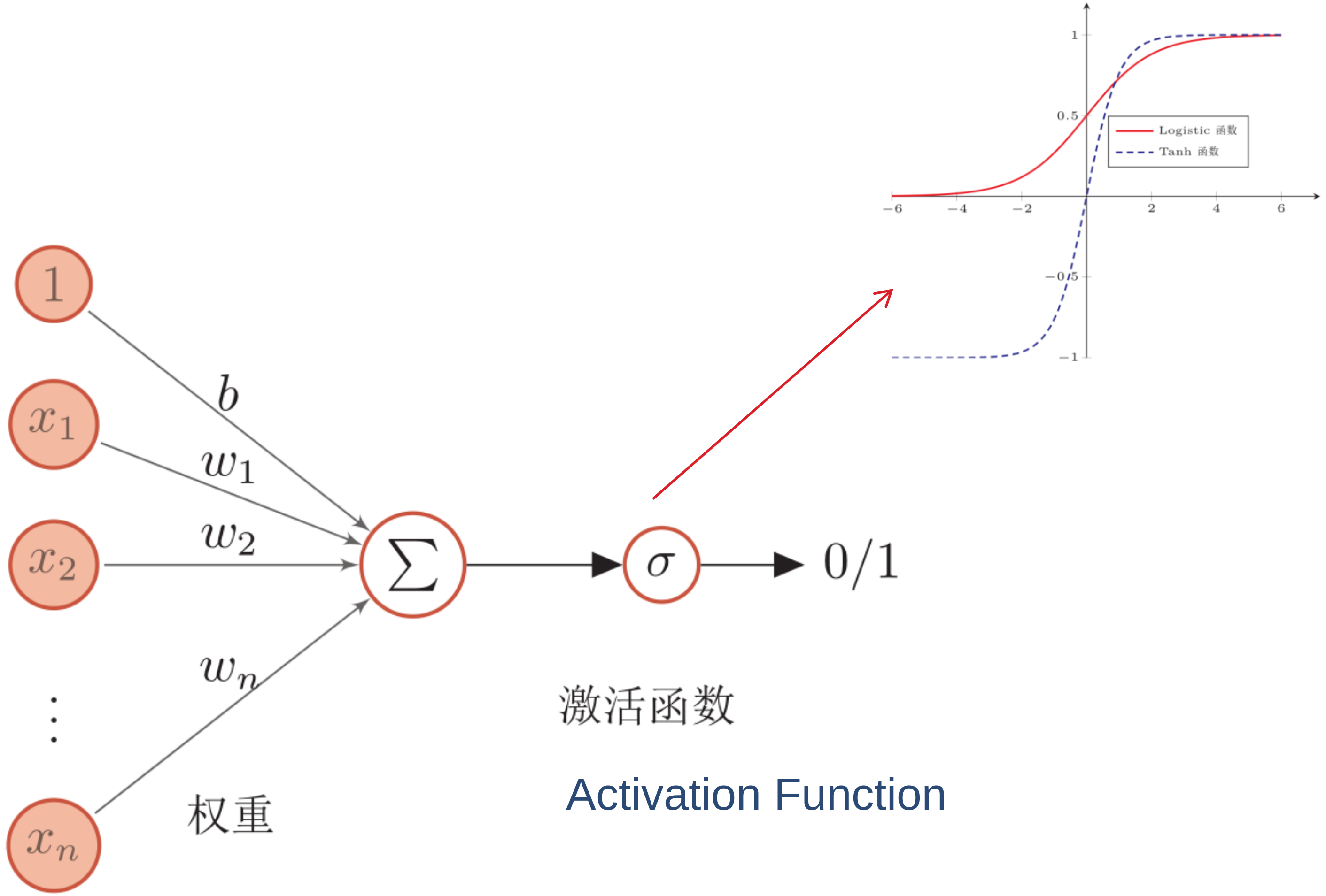
How to learn neural networks?

- **Hebb's Rule (赫布法则)**
  - When an axon of neuron A is close enough to neuron B to influence it and continuously and repeatedly participate in the excitement of neuron B, then some growth process or metabolic changes will occur in one or both of these neurons, so that neuron A, as one of the cells that can excite neuron B, has enhanced its efficiency.

----Canadian psychologist Donald Hebb
《Organization of behavior》 1949

- Memories in the human brain: long-term memory and short-term memory. Short-term memory lasts no more than one minute. If an experience is repeated enough times, it can be stored in long-term memory.
- The process of transforming short-term memory into long-term memory is called coagulation.
- Hippocampus in human brain is the core area of brain structure coagulation.

Artificial Neuron

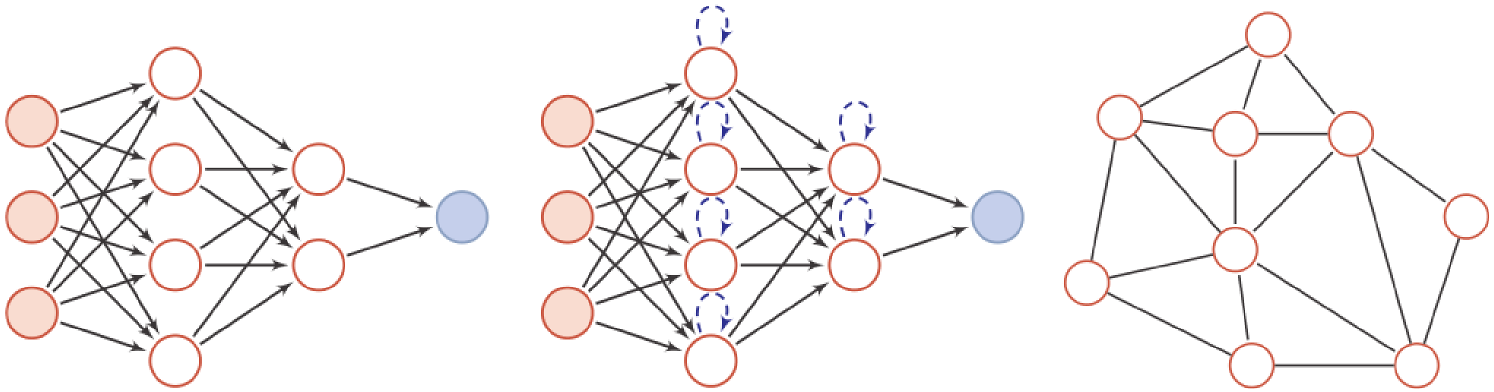


Artificial Neural Networks (ANNs)

- ANN is mainly composed of many neurons and their directed connections. So consider three aspects:
- Activation rules of neurons
  - mapping relationships between the input and output of neurons, which is generally a nonlinear function.
- Topological structures of networks
  - The connection between different neurons.
- Learning algorithms
  - The parameters of neural network are learned by training data.

Artificial Neural Networks (ANNs)

- ANN is composed of neuron: information processing network composed of many neurons has a parallel distribution structure.
- Here, the neural network is divided into three types:

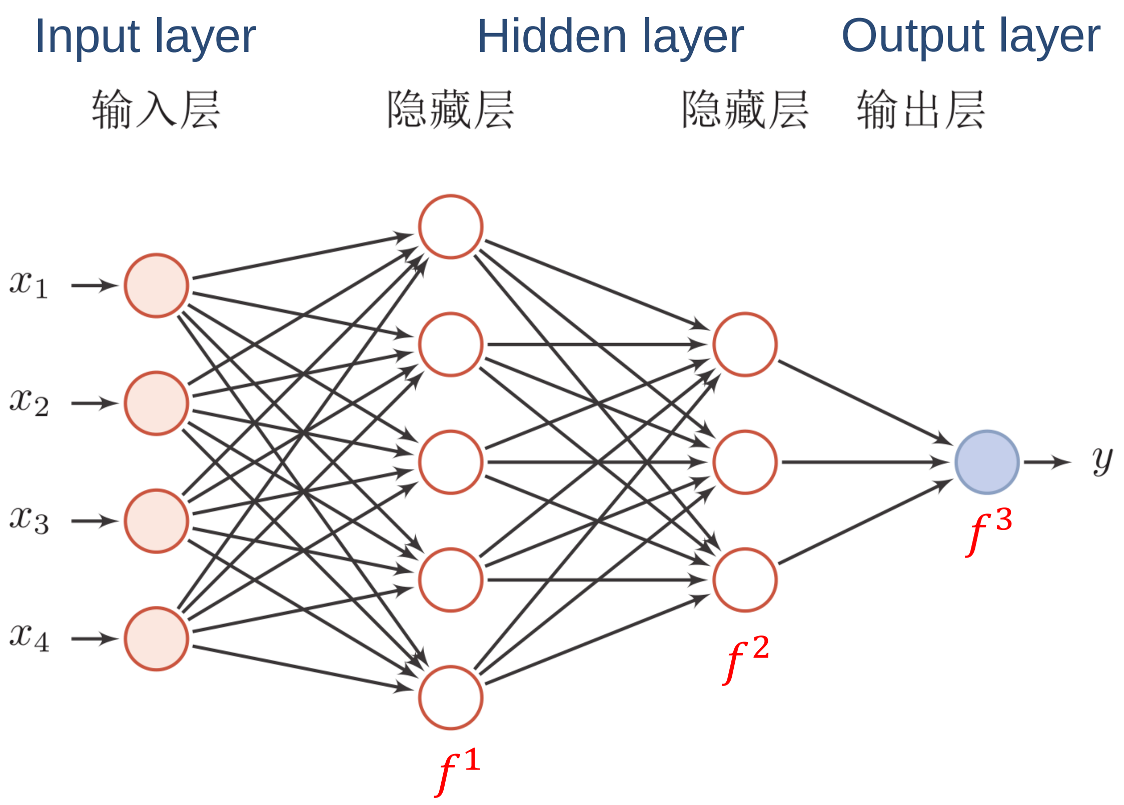


(a) 前馈网络 — Forward Network (FN)

(b) 记忆网络 — Memory Network

(c) 图网络 — Graph Network

神经网络

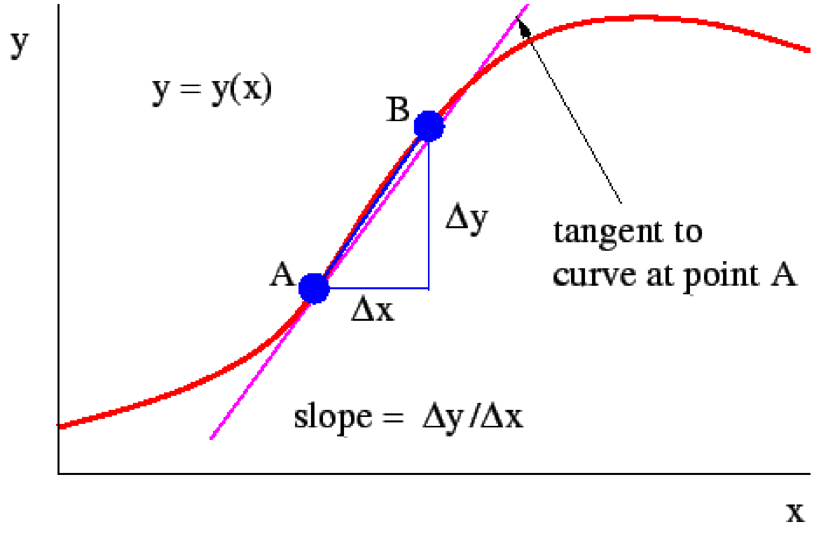


$$y = (f^3(f^2(f^1(x))))$$

$$f^l(x) = \sigma\left(W^{(l)} f^{l-1}(x)\right)$$

Contribution Distribution: Solved?

- Partial Derivative



- Contribution

$$\frac{\partial y}{\partial W^{(l)}} = \frac{y\left(W^{(l)} + \Delta W\right) - y(W^{(l)})}{\Delta W}$$

Neural network is not deep learning naturally, but deep learning is neural network naturally.

1.7 Development History of ML

History of Neural Networks (NNs)

- Development of NN has roughly gone through five stages
- First stage: model presentation
  - In 1943, psychologist Warren McCulloch and mathematician Walter Pitts first described an idealized ANN, constructed a calculation mechanism based on simple logical operations (MP model)
  - In 1948, Turing described a “B-type Turing Machine” (Hebb Learning)
  - In 1951, McCulloch and Pitts’ student Marvin Minsky built the first NN machine called SNARC.
  - In 1958, Rosenblatt first proposed a NN model that can simulate human perception, and called it “Perceptron”, and proposed a learning algorithm that is close to human learning process (iteration, trial and error).

History of Neural Networks (NNs)

Second stage: ice age (冰河期)

- In 1969, Marvin Minsky published "Perceptron”: NN were put into limbo directly, which led to the "ice age" of NN for more than ten years. They found two key problems of neural network:
  - The basic perceptron cannot handle XOR loops.
  - Computers are not capable enough to handle the long calculation time required by large-scale NNs.
- In 1974, Paul Webos (Harvard Uni.) invented the back propagation (BP) algorithm, but it was not paid due attention at that time.
- In 1980, Kunihiko Fukushima (福島邦彦) proposed a multi-layer neural network with convolution and sub-sampling operations: Neocognitron (新智机).

History of Neural Networks (NNs)

- Third stage: revival caused by BP algorithm
  - In 1983, physicist John Hopfield introduced the concept of energy function to NN, and proposed a network for associative memory and optimization calculation (Hopfield network), which obtained the best results at that time on the traveling salesman problem (TSP) and caused a sensation.
  - In 1984, Geoffrey Hinton proposed a randomized version of Hopfield network, namely Boltzmann machine.
  - In 1986, David Rumelhart and James McClelland provided a comprehensive discussion on the application of connectionism in computer simulation of neural activities, and re-invented the BP algorithm.

History of Neural Networks (NNs)

- Third stage: revival caused by BP algorithm
  - In 1986, Geoffrey Hinton and others will introduce the BP algorithm to the multilayer perceptron.
  - In 1989, LeCun and others introduced the BP algorithm into convolutional neural networks (CNNs), and achieved great success in handwritten numeral recognition.

History of Neural Networks (NNs)

- Fourth stage: the popularity is reduced
  - In the mid-1990s, statistical learning theory and machine learning model, such as support vector machine (SVM) began to rise.

This part, including statistical learning and ML
(not including NN) are called "traditional ML"
This is the core content of this course

  - In contrast, the shortcomings of NNs, such as unclear theoretical basis, difficult optimization and poor interpretability, are more prominent, and the research of NNs has once again fallen into a low tide.

History of Neural Networks (NNs)

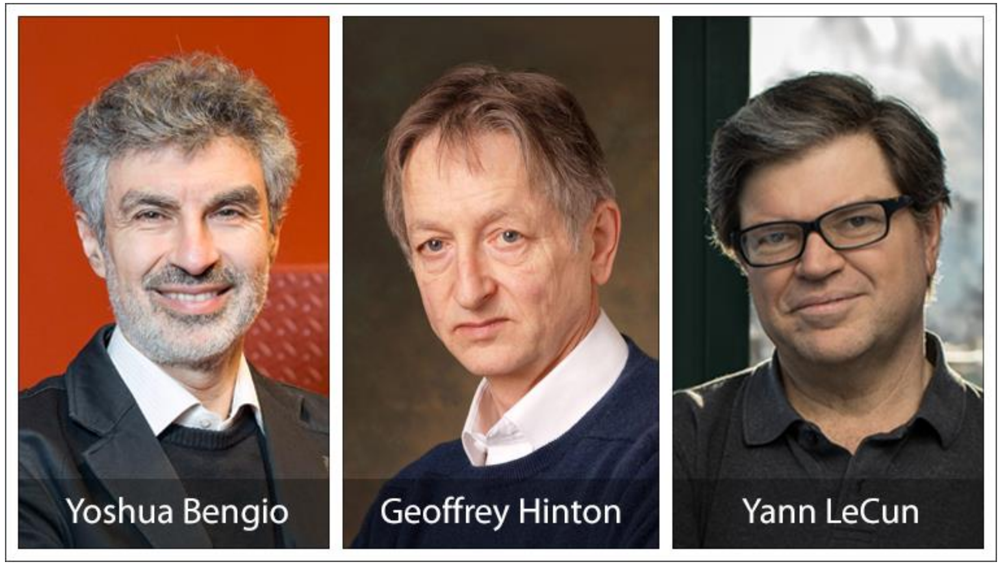
- Fifth stage: the rise of deep learning
  - In 2006, Hinton and others found that multi-layer feedforward NNs (MLP) can be effectively learned by pre-training layer by layer and then fine-tuning with BP algorithm.
    - Great success of deep NN in speech recognition and image classification.
  - In 2013, AlexNet: The first modern deep CNN was the beginning of a real breakthrough in image classification by deep learning.
    - AlexNet uses many modern deep network technologies for the first time without pre-training and layer-by-layer training.
    - With the large-scale parallel computing and popularity of GPU devices, the computing power of computers has been greatly improved. In addition, the scale of data available for ML is also growing. With the support of computing power and data scale, computers have been able to train large-scale ANNs.

Revolution of deep learning

- Artificial Intelligence (AI)
  - Speech recognition: It can reduce the word error rate from 1/4 to 1/8.
  - Computer vision: target recognition, image classification, etc.
  - Natural language processing: distributed representation, machine translation, question answering, etc.
  - Information retrieval and social network
- Three Deep：
  - Deep Blue
  - Deep QA
  - Deep Learning

Three carriages of Deep Learning

- Toronto University
  - Hinton: PhD of Edinburgh University in 1975
- NYU
  - Lecun (Now Facebook): Hinton's Postdoc in 1987
- Montreal University
  - Bengio: M. Jordan's Postdoc in 1991
- IDSIA
  - Jürgen Schmidhuber



2018 Turing award winner


AAAI-20
Thirty-Fourth AAAI Conference on Artificial Intelligence
February 7-12 2020, Hilton New York Midtown, New York, New York USA

AAAI 20 / AAAI 2020 Keynotes Turing Award Winners Event / Geoff Hinton, Yann Le Cunn, Yoshua Bengio

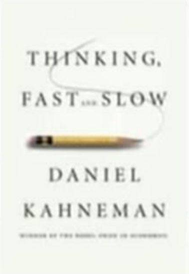
稍后观看 分享 信息

SYSTEM 1 VS. SYSTEM 2 COGNITION

2 systems (and categories of cognitive tasks):

Manipulates high-level / semantic concepts, which can be recombined combinatorially

System 1
- Intuitive, fast, **UNCONSCIOUS**, non-linguistic, habitual
- Current DL



System 2
- Slow, logical, sequential, **CONSCIOUS**, linguistic, algorithmic, planning, reasoning
- Future DL

Mila

humans reason not necessarily machines
as we train as we program them and and

1:13:41 / 2:10:16